



Cisco Firepower Next-Generation Firewall 6.6 Advanced Lab v1.6

Last Updated: 16-Oct-2020

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About This Demonstration

This guide for this preconfigured demonstration includes:

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Requirements

The table below outlines the requirements for this preconfigured demonstration.

Table 1. Requirements

Required	Optional
• Laptop	• Cisco AnyConnect®

About This Solution

IT teams have been asked to manage security using a patchwork of siloed point products, starting with legacy next-generation firewalls (NGFW), which were created with a focus on application and bolted on best effort threat protection. As such, these legacy NGFWs are unable to provide an enterprise with the contextual information, automation, and prioritization that they need to handle today's modern threats.

Cisco Firepower is an integrated suite of network security and traffic management products, deployed either on purpose-built platforms or as a software solution. The system is designed to help you handle network traffic in a way that complies with your organization's security policy-your guidelines for protecting your network.

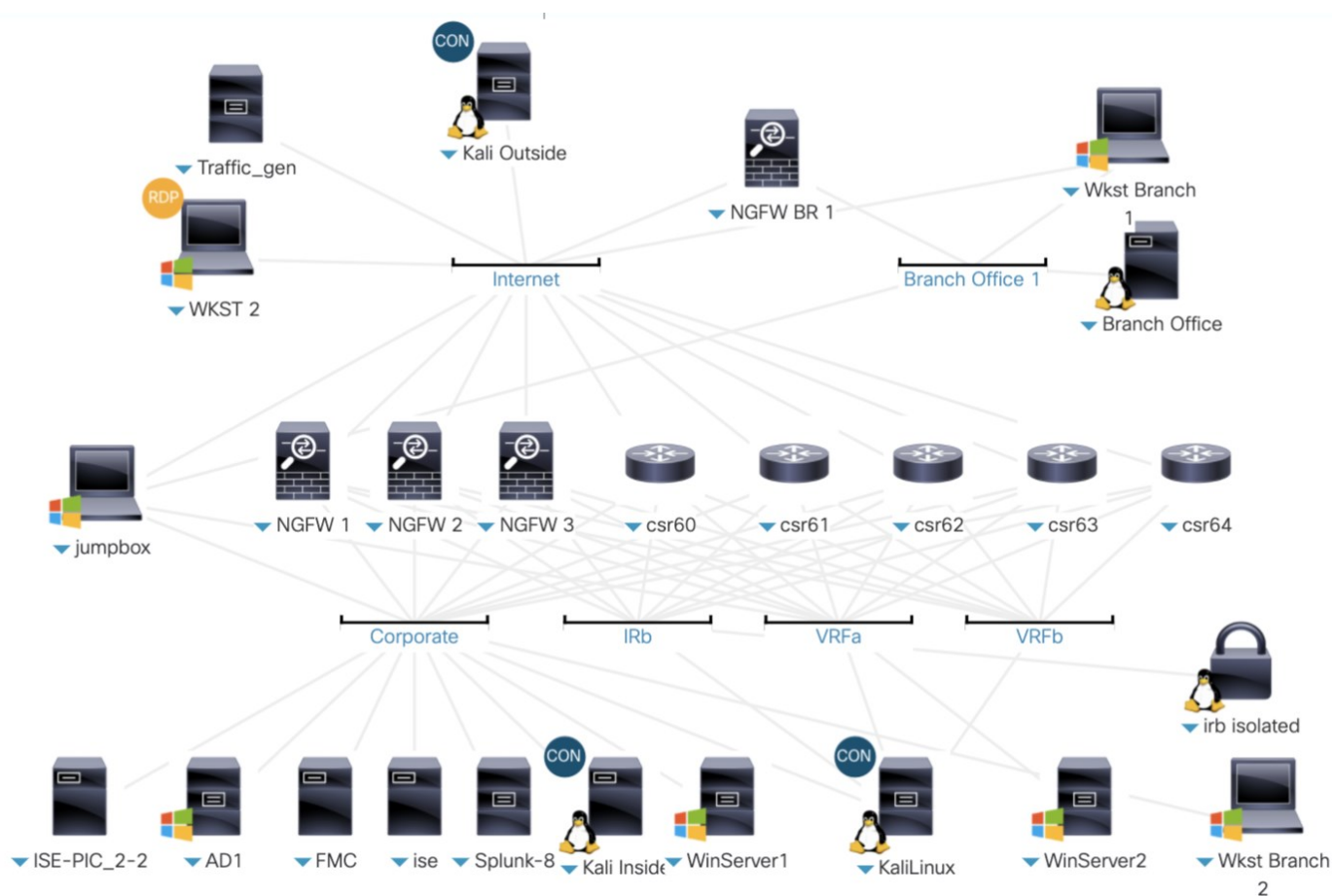
This allows the Cisco Firepower NGFW to evolve with a focus on enabling enterprises to stop, prioritize, understand, and automate responses to modern threats in real-time. Firepower NGFW is unique in its threat-focus, with a foundation of comprehensive network visibility, best-of-breed threat intelligence and highly effective threat prevention to address both known and unknown threats. Firepower NGFW also enables retrospective security, through Advanced Malware Protection, that can "go back in time" to quickly find and remediate sophisticated attacks that may have slipped through defenses. This has led to a significant reduction in time-to-detection (TTD) for Cisco customers compared to industry averages.

In this lab you will build a multi-site network Next Generation Firewall (NGFW) solution at between a corporate and two branch sites. Using the Firepower Management Console (FMC) you will build High Availability NGFWs at the corporate site and manage a branch. In this lab you will also configure a NGFW using the FDM (Firepower Device Manager). You will also configure remote access and site to site VPNs. You will also configure Cisco Threat Intelligence Director to accept and implement third party updates to your NGFW devices.

Topology

This content includes preconfigured users and components to illustrate the scripted scenarios and features of the solution. Most components are fully configurable with predefined administrative user accounts. You can see the IP address and user account credentials to use to access a component by clicking the component icon in the **Topology** menu of your active session and in the scenario steps that require their use.

Figure 1. dCloud Topology



Get Started

BEFORE PRESENTING

Cisco dCloud strongly recommends that you perform the tasks in this document with an active session before presenting in front of a live audience. This will allow you to become familiar with the structure of the document and content.

It may be necessary to schedule a new session after following this guide in order to reset the environment to its original configuration.

PREPARATION IS KEY TO A SUCCESSFUL PRESENTATION.

Follow the steps to schedule a session of the content and configure your presentation environment.

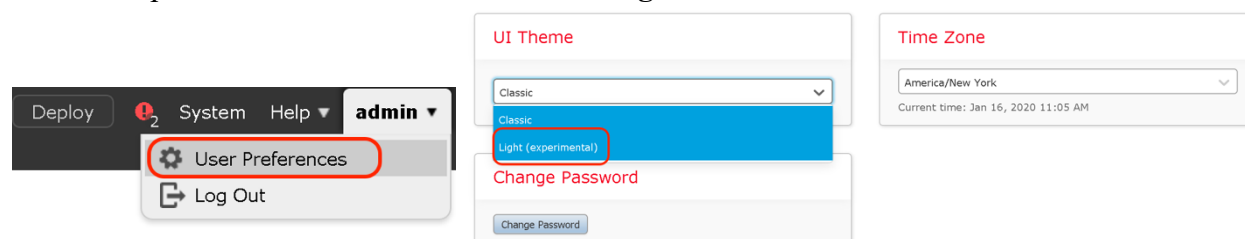
1. Initiate your dCloud session. [[Show Me How](#)]

NOTE: It may take up to 10 minutes for your session to become active.

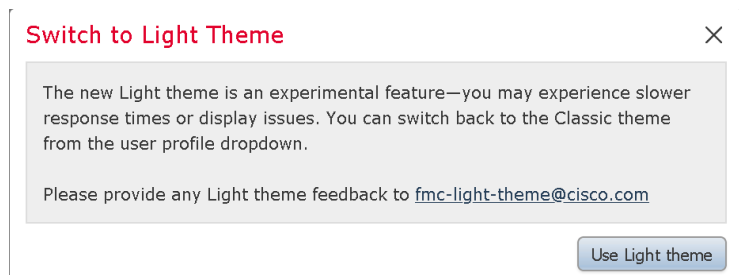
2. For best performance, connect to the workstation with **Cisco AnyConnect VPN** [[Show Me How](#)] and the **local RDP client on your laptop** [[Show Me How](#)]

Jump PC: **198.18.133.50**, Username: **administrator**, Password: **C1sco12345**

3. From the jumpbox, open **Firefox**. It should open with FMC login page as the default page. The login name and password will be prepopulated. Click **Login**
4. Navigate to **admin > User Preferences** in the top right-hand corner. Under the **General** tab, click on the dropdown under **UI Theme** and select **Light** theme.



5. A popup will show up with a disclaimer about the Light Theme. Click on **Use Light theme**.



6. The FMC UI will change from the classic UI to new UI.

UI Theme

Light (experimental) ▼

Please provide any Light theme feedback to
fmc-light-theme@cisco.com

Time Zone

America/New York ▼

Current time: Jan 16, 2020 11:09 AM

Change Password

Change Password

The above step is optional for Firepower release 6.5. In Firepower release 6.6, the above UI theme is used by default. However, all the lab screenshots were captured using the new UI theme. It has no impact on the configuration steps.

NOTE: You can also connect to the workstation using the Cisco dCloud Remote Desktop client [[Show Me How](#)]. The dCloud Remote Desktop client works best for accessing an active session with minimal interaction. However, many users experience connection and performance issues with this method. Check the connection for Remote Desktops for Wkstbr2 make sure you get the Login prompt password C1sco12345

Scenario 1. Integrated Routing and Bridging (IRB)

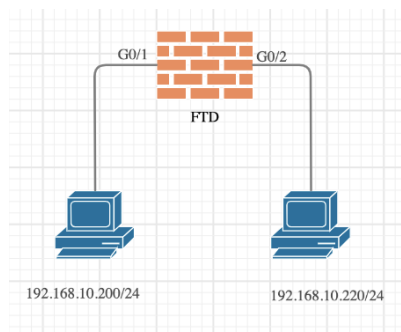
The Integrated Routing and Switching (IRB) feature allows users to configure a bridge group interface on the FTD even in the routed mode. With this feature, the bridge group interfaces can co-exist with L3 interfaces as well. Unlike, Bridge groups in Transparent mode, communication across different BVI interfaces is supported in routed mode. The configuration for this feature is done via the FMC.

This exercise consists of the following tasks.

- Create the objects needed for this lab exercise
- Modify the NGFW interface configuration
- Modify the NAT policy
- Modify the access control policy
- Deploy and test the configuration

In the lab, there is a Linux server on separate VLAN that is connected to GigabitEthernet0/2. The FQDN for this server **isolated.dcloud.local**, and it has the IP address of 198.19.10.220/24. Note that this address is in the same subnet as the inside network.

The objective is to join these VLANs using a bridge-group on the NGFW. Traffic between these VLANs will be inspected.



NOTE: In this exercise, both interfaces in the bridge group are put in same security zones. However, a bridge group can contain interfaces in different security zones. Putting the bridge-group interfaces in different zones allows more granular control of traffic between interfaces in the same bridge group.

Steps

Configure the Interface Security Zone object

Verify the objects needed for this lab exercise:

The screenshot shows the Firepower Management Center (FMC) interface. The 'Objects' tab is selected, and the 'Interface' sub-tab is active. A dropdown menu is open under 'Object Management', showing 'Object Management' and 'Intrusion Rules'. The 'Interface' table shows a single entry: 'BVIZone' with Domain 'Global', Type 'Security Zone', and Interface Type 'Switched'.

1. Open **Firefox** and open the **Firepower Management Center** (labeled FMC). The login name and password will prepopulate.
2. Click **Log In**.
3. Navigate to **Objects > Object Management**. Select **Interface** from the left navigation panel.
4. Click **Add > Security Zone**.
 - a. For Name, enter **BVIZone**.
 - b. Select **Switched** from the Interface Type drop-down menu.
 - c. Click **Save**

NOTE: The type 'switched' is used to change the interface to L2 software switched mode when this zone is assigned to the member interfaces of the BVI.

Modify the NGFW interface configuration

Here we are creating BVI interface with G0/1 and G0/2 as its members.

1. Navigate to **Devices > Device Management > NGFW1**.

The screenshot shows the Firepower Management Center (FMC) interface. The 'Devices' tab is selected. A table lists two devices: 'NGFW1' and 'NGFWBR1'. The 'NGFW1' device is highlighted, and its configuration is shown in the right pane. The 'Interfaces' tab is selected, and the 'Add Interfaces' button is visible.

- a. Click on the pencil icon to edit the NGFW1 device configuration, and select the **Interfaces** tab.
- b. Click **Add Interfaces** and select **Bridge Group Interface**.
- c. For **Name** enter **InsideBVI**.

Add Bridge Group Interface

Interfaces IPv4 IPv6

Name: InsideBVI

Description:

Bridge Group ID *: 1

Available Interfaces

Search

- GigabitEthernet0/0
- GigabitEthernet0/1
- GigabitEthernet0/2**

Selected Interfaces

- GigabitEthernet0/1
- GigabitEthernet0/2

Add

OK Cancel

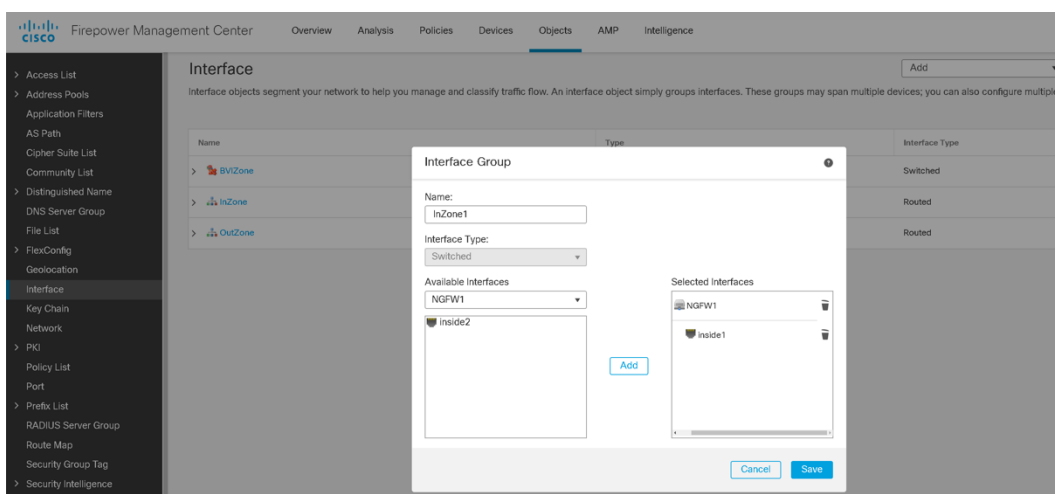
- d. For Bridge Group ID, enter **1**
- e. Select **GigabitEthernet0/1** and **GigabitEthernet0/2** and click **Add**.
- f. Select the IPv4 tab and enter the IP address **198.19.10.1/24**.
- g. Click **OK**.
 - i. Please take a moment to read the information in the popup box
 - ii. Click **Yes**

NOTE: Zone assignments and IP address configuration will be removed from GigabitEthernet0/1 and GigabitEthernet0/2 when they are added as members of a Bridge Group interface.

2. Click on the pencil icon to edit the **GigabitEthernet0/1** interface.
 - a. For Name enter **inside1**.
 - b. Confirm that the Enabled checkbox is checked.
 - c. Select **BVIZone** from the Security Zone drop-down list.
 - d. Click **OK**.
3. Click on the pencil icon to edit the **GigabitEthernet0/2** interface.
 - a. For Name enter **inside2**.
 - b. Check the **Enabled** checkbox.
 - c. Select **BVIZone** from the Security Zone drop-down list.
 - d. Click **OK**.
4. Click **Save** to save the device configuration.

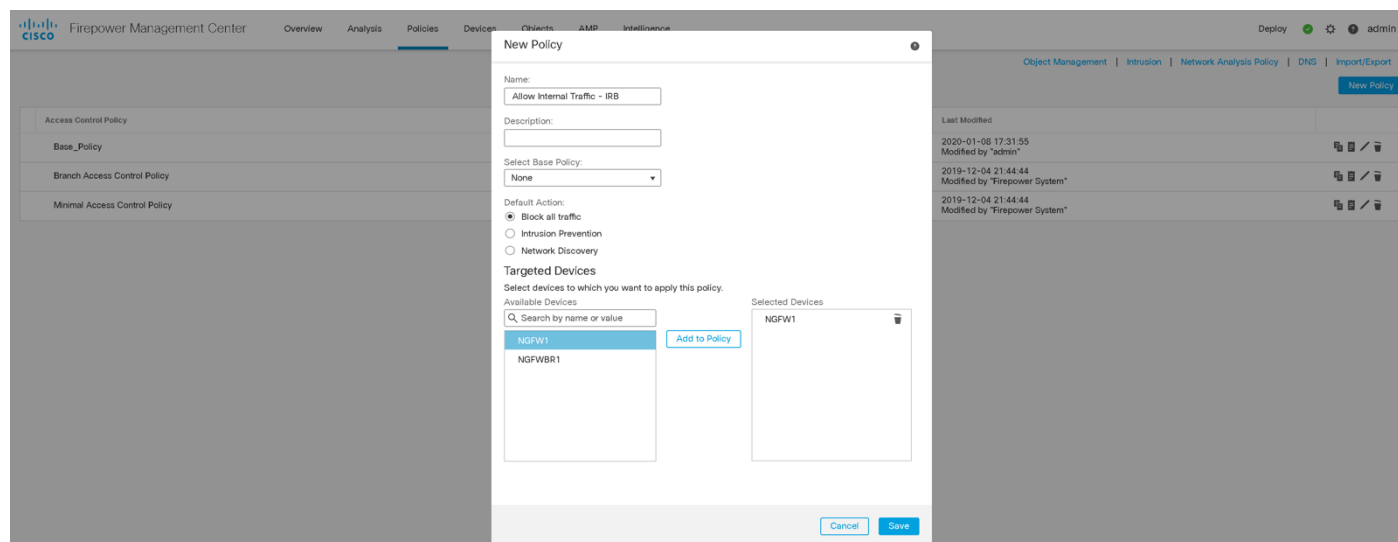
Modify the NAT policy

NOTE: If you want the static NAT rule to work with the BVI interfaces, you must include this step. This is because object NAT does not allow zones with more than one interface.

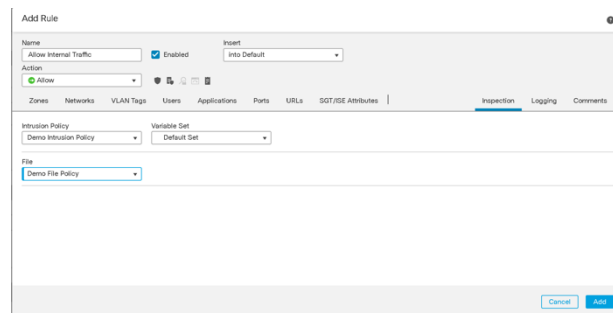
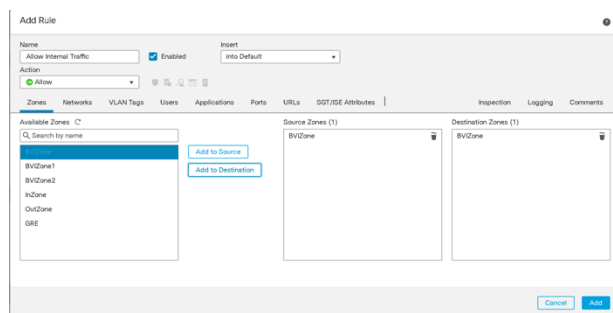


1. Navigate to **Objects > Object Management**. Select **Interface** from the left navigation panel.
 - a. Click **Add > Interface Group**.
 - b. For **NAME**, enter **InGroup1**.
 - c. For **Interface Type**, select **Switched**.
 - d. Select the interface **inside1** and click **Add**.
 - e. Click **Save**.
2. Navigate to **Devices > NAT**.
3. Edit the **Default_PAT** policy.
 - a. Replace **InZone1** with **InGroup1** in all the **Auto NAT** rules.
 - b. Replace **InZone1** with **BVIZone** in every other rule.
 - c. Click **Save** to save the NAT policy.

Create an access control policy for inspecting traffic between BVI interfaces



1. Create a new Access Policy Named **Allow Internal Traffic – IRB**
 - a. Navigate to **Policies > Access Control > Access Control**.
 - b. Click on the **New Policy** button and name it **Allow Internal Traffic – IRB**
 - c. Select Base Policy as **None** and Default Action as **Intrusion Prevention**
 - d. Select **NGFW1** from the targeted devices
 - e. Click **Save**
 - f. A prompt will be displayed stating that the policy assignment for the **NGFW1** will change from **Base_Policy**
 - g. Click **Yes**



2. Add an access control rule to allow (but inspect) traffic between interfaces in **BVIZone**.
 - a. For **Name**, enter **Allow Internal Traffic**.
 - b. Select **into Default rule** from the **Insert** drop-down list
 - c. The **Zones** tab should already be selected.
 - d. Select **BVIZone** and click **Add to Source**.
 - e. Select **BVIZone** and click **Add to Destination**.
 - f. Select the **Inspection** tab.

- g. Select **Demo Intrusion Policy** from the Intrusion Policy drop-down list.
- h. Select **Demo File Policy** from the File Policy drop-down list.
- i. Click **Add** to add the rule.
3. Click **Save** to save the changes to the access control policy.
4. Deploy the policies to NGFW1 and test the configuration.

Testing the IRB Policy Configuration

NOTE: Deploy the configuration changes and wait for the deployment to complete.

1. From the **Inside Linux Server** [root/C1sco12345] CLI, test connectivity by typing **ping isolated**. This should succeed.
 - a. If the ping does not succeed check the Mandatory rules for the Base_Policy and change the Destination Zone from any to a different zone.
2. From the Inside Linux Server CLI, test the IPS capabilities.
 - a. Run the following command from the Inside Linux server CLI. **ftp isolated**
 - b. Login as **guest**, password **C1sco12345**
 - i. Type **cd ~root**. You should see the following message:
 - ii. **421 Service not available, remote server has closed connection**
 - c. From the Inside Linux server CLI, test the file and malware blocking capabilities.
 - i. As a control test, use WGET to download a file that is not blocked. **wget -t 1 isolated/files/ProjectX.pdf**. This should succeed.
 - ii. Next use WGET to attempt to download the file blocked by type. **wget -t 1 isolated/files/test3.avi**

NOTE: Very little of the file is downloaded. This is because the NGFW can detect the file type when it sees the first block of data. The *Demo File Policy* is configured to block AVI files.

- iii. Finally use WGET to attempt to download malware. **wget -t 1 isolated/files/Zombies.pdf**

NOTE: About 99% of the file is downloaded. This is because the NGFW needs the entire file to calculate the SHA. The NGFW holds onto the last block of data until the hash is calculated and looked up. The *Demo File Policy* is configured to block malware detected in PDF files.

Remove IRB Lab components

1. **Devices > Device Management >** click the **pencil** icon on the **NGFW1** line
2. Click **Remove Icon** from the **BVI1 Interface**
3. Go to **GigabitEthernet 0/1** and click the **pencil** icon
 - a. Name **in10**
 - b. Click **Enabled**

- c. Security Zone **InZone1**
 - d. IPv4 **198.19.10.1/24**
 - e. Click **OK**
4. Go to GigabitEthernet 0/2 and remove the name from the interface and make sure it is enabled.
- a. Name **in20**
 - b. Click **Enabled**
 - c. Security Zone **InZone2**
 - d. IPv4 **198.19.20.1/24**
 - e. Click **OK**
5. Go to Device NAT **Default PAT**
- a. Replace **InGroup1** with **InZone1** for all NAT Rules
 - b. Replace the **BVIZone** with **InZone1**
 - c. Click **Save**
6. Go to **Policies > Access Control > Base_Policy**
- a. Click the **pencil** icon
 - b. Replace **ALL BVIZone** with **InZone1**
 - c. Delete the Block ICMP over GRE line 1 in Mandatory
 - d. You can delete the line with **BVIZone to BVIZone used for the Allow Internal Traffic**
 - e. You can delete the line with Allow GRE Traffic also
 - f. Click **Save**
7. Click **Deploy**
- a. Select NGFW1

Test the Connectivity after removing IRB configuration

1. From **Inside Linux Server** [root/C1sco12345]
 - a. Ping **Outside**
2. From **Outside Linux Server** [root/C1sco12345]
 - a. **Ping 198.18.133.120** (Outside NAT Address of FMC)
 - b. **Ping 198.18.128.202** (Outside NAT Address of Inside Linux Server)
 - c. Open a putty session to NGFWBR1 Username: admin Password: C1sco12345
 - i. Open the Command Prompt
 1. **Ping 198.18.133.120** (Outside NAT Address of FMC)
 2. **Ping 198.18.128.202** (Outside NAT Address of Inside Linux Server)

Scenario 2. High Availability Configuration

This exercise consists of the following tasks.

- Configure and Deploy Backup NGFW
- Create High Availability Pair of Firewalls
- Configure Active/Standby with Virtual Mac Address
- Test the configuration

The objective of this exercise is to understand and configure High Availability for NGFW. You will configure the second firewall and then add it to the High Availability group.

Steps

Onboard a backup NGFW onto the FMC

NGFW device onboarding onto the FMC is done in two parts. First, configuration settings are added on the NGFW to configure FMC as the device manager and establish communication with the FMC on the management interface. Secondly, the NGFW device is added onto the **Devices** page on the FMC to initiate and establish communication. Here, we use the **jumpbox** to complete both the steps.

1. For configuring the backup NGFW, go to the Windows machine labelled **jumpbox** and open the PUTTY Session - Select **NGFW3**, Select **Load** and Select **Open**
 - a. Username: **admin** password: **C1sco12345**
 - b. Type: **show managers**
 - c. If it says **Managed locally** type:
 - a **configure manager delete** and type **yes**
 - b Now, the output of **show managers** will show **No managers configured**. then follow sub-steps in [d] below
 - d. If it says **No managers configured**
 - i. Type the following: **configure manager add fmc.dcloud.local C1sco12345** select **yes**, if prompted (must type yes in full)
 - ii. When command prompt returns type: **show managers** make sure fmc.dcloud.local shows **“status pending”**

```

NGFW3
login as: admin
Using keyboard-interactive authentication.
Password:
Last login: Thu Jan  9 15:21:01 UTC 2020 from jump.dcloud.local on pts/0

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Cisco Fire Linux OS v6.5.0 (build 4)
Cisco Firepower Threat Defense for VMWare v6.5.0 (build 115)

> show managers
Managed locally.

> configure manager delete

If you enabled any feature licenses, you must disable them in Firepower Device Manager before deleting the local manager.
Otherwise, those licenses remain assigned to the device in Cisco Smart Software Manager.
Do you want to continue[yes/no]:yes

>
> show managers
No managers configured.

> configure manager add fmc.dcloud.local Cisco12345
Manager successfully configured.
Please make note of reg_key as this will be required while adding Device in FMC.

> show managers
Host                : fmc.dcloud.local
Registration Key     : ****
Registration        : pending
RPC Status          :

```

NOTE: The NGFW can be added to the **FMC for management** using the FMC UI. However, here we are going to use REST API to demonstrate the automated capabilities of the solution. A python script is present on the **Inside Linux Server** called **runapiscript** at the location **/usr/local/bin**. It invokes the REST APIs on the FMC to register the NGFW3 with the FMC, applies the Health Policy to the NGFW3, discovers the interfaces, configures the IP addresses and deploys the Access Control Policy with the name specified during the execution of the script.

2. For adding the backup NGFW to the FMC, open the PUTTY Session and Select **Inside Linux server**, Select **Load** and Select **Open**
 - a. Login using the credentials - **[root/Cisco12345]**
 - b. Type **runapiscript** wait for the prompt
 - c. When asked **Which Firewall do you want to register?** Type the number **3**
 - d. When it asked **Enter name of new Access Control Policy to be create**, type **HA** for the name

```

root@inside:~
[root@inside ~]# runapiscript
Specify firewall to register.
  Enter 1 for NGFW1.
  Enter 2 for NGFW2.
  Enter 3 for NGFW3.
Which firewall do you want to register? 3
Enter name of new Access Control Policy to be create: HA
status code is: 201
Post was sucessfull...
Access Control Policy HA created.

Attempting to register NGFW3.
status code is: 202
Post was sucessfull...
Registration is in progress. (1)
Registration is in progress. (2)
Registration is in progress. (3)
Registration is in progress. (4)
Registration is in progress. (5)
Registration is in progress. (6)
Registration is in progress. (7)
Registration is in progress. (8)
Registration is in progress. (9)
Registration completed.

status code is: 200
GET was sucessfull...
Device found; setting ID.
Interface discovery is in progress. (1)
Interface discovery is in progress. (2)
Interface discovery completed.

status code is: 200
GET was sucessfull...
interface 1 found
interface 2 found
Attempting to set outside interface IP address to 198.18.133.83/18.
status code is: 200
Put was sucessfull...
Attempting to set inside interface IP address to 198.19.10.3/24.
status code is: 200
Put was sucessfull...
[root@inside ~]#

```

Note: The ‘Registration is in progress’ count may vary during the execution of the script.

3. Verify that the NGFW3 is added successfully to the FMC by checking the registration status of **NGFW3** on the FMC. You might need to allow some time for the device to register. It should show a **Green** symbol in front of the device name.

Device Management

List of all the devices currently registered on the Firepower Management Center

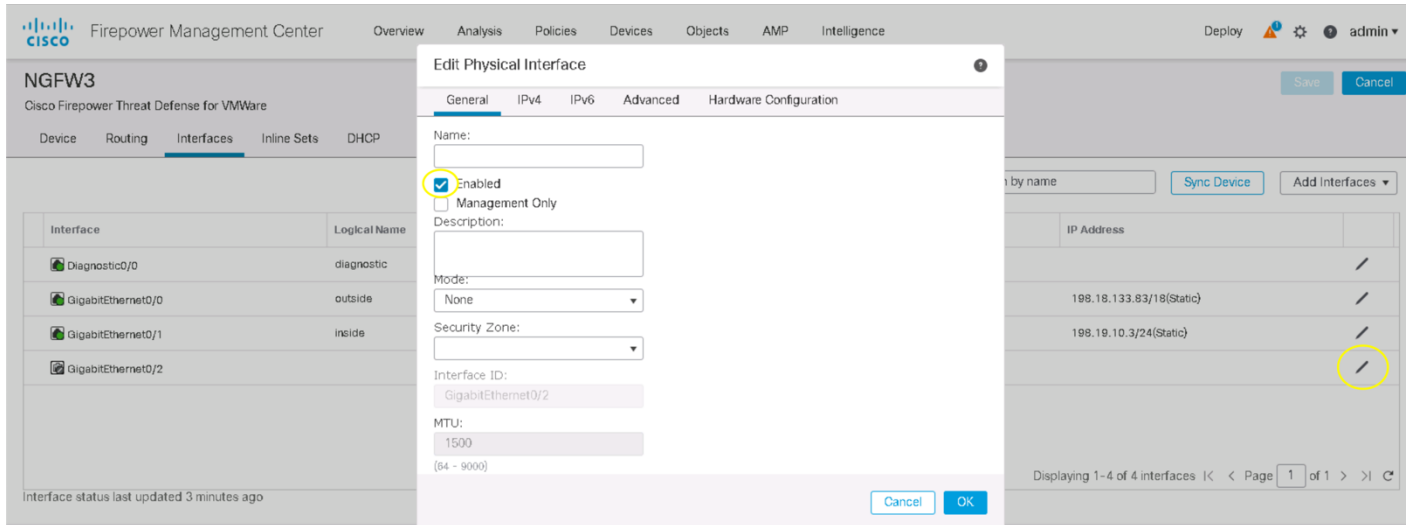
View By: Group

Legend: All (3) | Error (0) | Warning (0) | Offline (0) | Normal (3) | Deployment Pending (3)

Name	Model	Version	Chassis	Licenses	Access Control Policy
NGFW1 198.19.10.81 - Routed	FTD for VMWare	6.5.0	N/A	Base, Threat (4 more...)	Base_Policy
NGFW3 198.19.10.83 - Routed	FTD for VMWare	6.5.0	N/A	Base, Threat (2 more...)	HA
NGFWBR1 198.18.133.42 - Routed	FTD for VMWare	6.5.0	N/A	Base, Threat (2 more...)	Branch Access Control Policy

4. Verify the GigabitEthernet0/2 interface for NGFW3 is set to blank configuration. This interface is to be used as the failover interface during the formation of the HA pair.
 - a. Click on the **Pencil** icon next to NGFW3
 - b. Click on the **Pencil** icon next to the interface **GigabitEthernet0/2**

- c. Select the checkbox against **Enabled**
- d. **Delete** the **Name** of the interface (if any)
- e. Click **Ok** and then **Save**.
- f. Click **Deploy FMC UI** to push the interface state change to NGFW3



NOTE: The following information is communicated over the failover link:

The unit state (active or standby)

Hello messages (keep-alives)

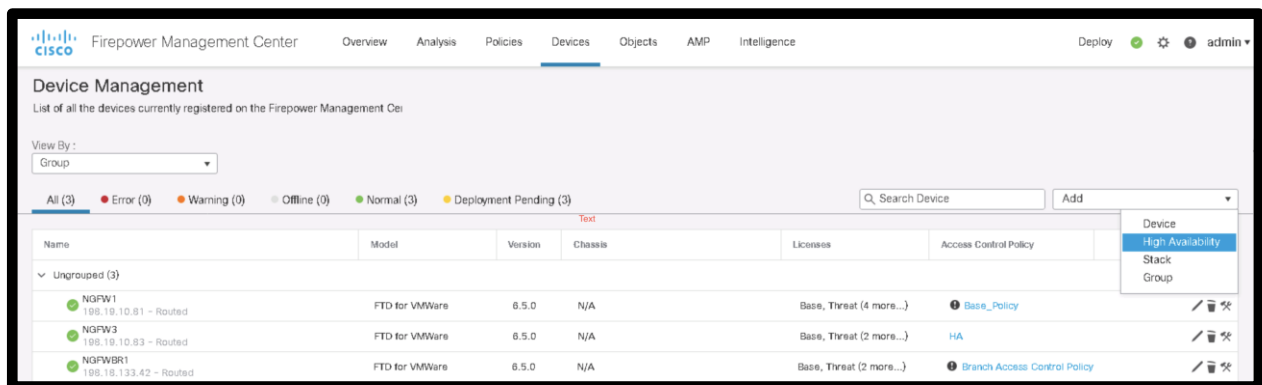
Network link status

MAC address exchange

Configuration replication and synchronization

Configure High Availability Pair

1. Go to **Devices > Device Management> Add > Add High Availability**



NOTE: The NGFW3 Management Interface (198.19.10.83) was preconfigured. Interfaces G0/0 and G0/1 were configured by the REST API script. They do not have security zones listed on the interface, but they will inherit the security zones and the interface IP Address' from NGFW1 when the HA process is run.

- a Name: **HA_TEST**
- b Device Type: **Firepower Threat Defense** (This is the device type associated with the NGFW)
- c Primary Peer: **NGFW1**
- d Secondary Peer: **NGFW3**
- e Then **Continue**
- f The formation of HA pair restarts the snort engine on both the Primary and Secondary NGFW devices. This is entirely expected behavior. Select **Yes**

Add High Availability Pair

Name:*

Device Type:

Primary Peer:

Secondary Peer:

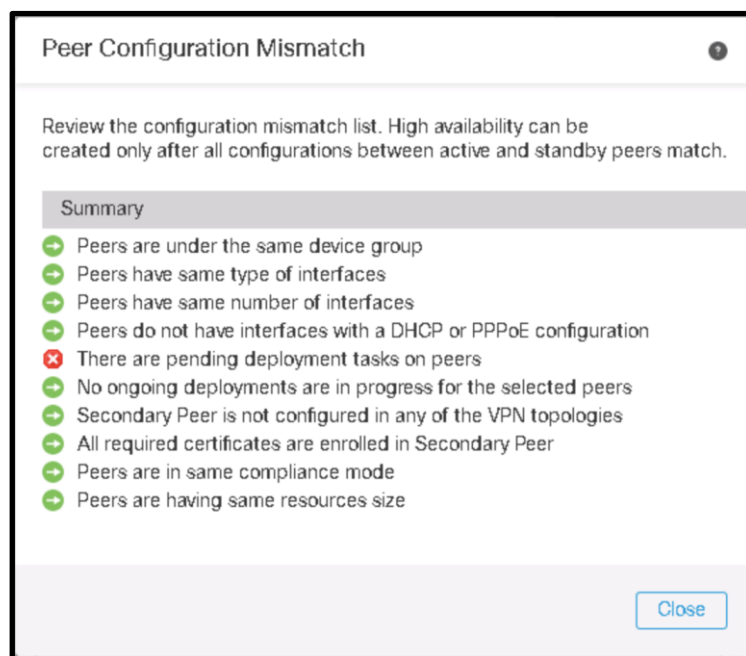
i Threat Defense High Availability pair will have primary configuration. Licenses from primary peer will be converted to their high availability versions and applied on both peers.

Warning

 This operation restarts the Snort processes of primary and secondary devices, temporarily causing traffic interruption.

Do you want to continue?

☐ Do not display this message again



NOTE: If you have done configuration tasks on either of the HA Peers and have not deployed then you will get the above message. Click on **Close** and then click **Cancel**.

2. Deploy the changes to **NGFW3** by clicking the **Deploy** button in the FMC.
3. Go back and repeat step 1
4. Now, enter the High Availability details as below :
 - a. Select Interface: GigabitEthernet0/2
 - b. Name: Failover_Link
 - c. Primary IP: 198.19.254.1
 - d. Secondary IP: 198.19.254.2 Subnet Mask: 255.255.255.0
 - e. State Link: Interface Same as LAN Failover
 - f. IPsec Encryption: Enabled (OPTIONAL)
 - g. Click **Add**

NOTE: If Interfaces do not show up go back to Devices > Device Manager > Click on the Pencil Icon for each firewall click on the Interfaces to make sure that the interfaces do not have names.

Add High Availability Pair

High Availability Link

Interface:* GigabitEthernet0/2 ▼

Logical Name:* Failover_Link

Primary IP:* 198.19.254.1

☐ Use IPv6 Address

Secondary IP:* 198.19.254.2

Subnet Mask:* 255.255.255.0

State Link

Interface:* Same as LAN Failover ▼

Logical Name:* Failover_Link

Primary IP:* 198.19.254.1

☐ Use IPv6 Address

Secondary IP:* 198.19.254.2

Subnet Mask:* 255.255.255.0

IPsec Encryption

☒ Enabled

Key Generation: Auto ▼

i LAN failover link is used to sync configuration, stateful failover link is used to sync application content between peers. Selected interface links and encryption settings cannot be changed later.

Cancel
Add

NOTE: The Failover ‘State link’ can be a separate dedicated link as well. However, for simplicity in this lab we are using the LAN failover link as the State link also.

Creating or breaking a Firepower Threat Defense high availability pair immediately restarts the Snort process on the primary and secondary devices, temporarily interrupting traffic inspection on both devices. Whether traffic drops during this interruption or passes without further inspection depends on the model of the managed device and how it handles traffic. See Snort® Restart Traffic Behavior for more information. The system warns you that continuing to create a high availability pair restarts the Snort process on the primary and secondary devices and allows you to cancel.

5. Click on OK to add the High Availability Pair

High Availability Pair Added



Configure interface monitoring options to protect the HA pair interfaces. Apply the same health policies to the primary and secondary nodes of the HA pair.



NOTE: The configuration of the HA will take some time you will see status updates from time to time if you watch the Tasks next to the deployment button.

Name	Model	Version	Chassis	Licenses	Access Control Policy	
Ungrouped (2)						
HA_TEST High Availability						
NGFW1(Primary, Unknown) 198.19.10.81 - Routed	FTD for VMWare	6.5.0	N/A	Base, Threat (4 more...)	Base_Policy	
NGFW3(Secondary, Unknown) 198.19.10.83 - Routed	FTD for VMWare	6.5.0	N/A	Base, Threat (4 more...)	Base_Policy	

6. When complete you will see the following:

Cisco Firepower Management Center						
Overview Analysis Policies Devices Objects AMP Intelligence						
Deploy admin						
Device Management						
List of all the devices currently registered on the Firepower Management Center						
View By: Group						
All (3) Error (0) Warning (0) Offline (0) Normal (3) Deployment Pending (1)						
Search Device Add						
Ungrouped (2)						
HA_Test High Availability						
NGFW1(Primary, Active) 198.19.10.81 - Routed	FTD for VMWare	6.5.0	N/A	Base, Threat (4 more...)	Base_Policy	
NGFW3(Secondary, Standby) 198.19.10.83 - Routed	FTD for VMWare	6.5.0	N/A	Base, Threat (4 more...)	Base_Policy	
NGFWBR1 198.18.133.42 - Routed	FTD for VMWare	6.5.0	N/A	Base, Threat (2 more...)	Branch Access Control Policy	

7. Go to **Devices > Device Management** Click on the pencil icon next to the HA pair name - **HA_Test**

Name	Model	Version	Chassis	Licenses	Access Control Policy	
Ungrouped (2)						
HA_Test High Availability						
NGFW1(Primary, Active) 198.19.10.81 - Routed	FTD for VMWare	6.5.0	N/A	Base, Threat (4 more...)	Base_Policy	
NGFW3(Secondary, Standby) 198.19.10.83 - Routed	FTD for VMWare	6.5.0	N/A	Base, Threat (4 more...)	Base_Policy	
NGFWBR1 198.18.133.42 - Routed	FTD for VMWare	6.5.0	N/A	Base, Threat (2 more...)	Branch Access Control Policy	

HA_TEST

Cisco Firepower Threat Defense for VMWare

Save Cancel

Summary

High Availability

Device

Routing

Interfaces

Inline Sets

DHCP

High Availability Configuration

High Availability Link

Interface

GigabitEthernet0/2

Logical Name

Fallover_Link

Primary IP

198.19.254.1

Secondary IP

198.19.254.2

Subnet Mask

255.255.255.0

IPsec Encryption

Enabled

State Link

Interface

GigabitEthernet0/2

Logical Name

Fallover_Link

Primary IP

198.19.254.1

Secondary IP

198.19.254.2

Subnet Mask

255.255.255.0

Statistics

Q

Monitored Interfaces

Interface Name	Active IPv4	Standby IPv4	Active IPv6 - Standby IPv6	Active Link-Local IPv6	Standby Link-Local IPv6	Monitoring	
outside	198.18.133.61					+	/
diagnostic						+	/
inside	198.19.10.1					+	/

Fallover Trigger Criteria

Failure Limit

Failure of 1 Interfaces

Peer Poll Time

1 sec

Peer Hold Time

15 sec

Interface Poll Time

5 sec

Interface Hold Time

25 sec

Interface MAC Addresses

Physical Interface

Active Mac Address

Standby Mac Address

No records to display

NOTE: MAC Addresses and IP Addresses in Failover.

When you configure your interfaces, you can specify an active IP address and a standby IP address on the same network.

Although recommended, the standby address is not required. Without a standby IP address, the active unit cannot perform network tests to check the standby interface health; it can only track the link state. You also cannot connect to the standby unit on that interface for management purposes.

When the primary unit or failover group fails over, the secondary unit assumes the IP addresses and MAC addresses of the primary unit and begins passing traffic.

The unit that is now in standby state takes over the standby IP addresses and MAC addresses.

Because network devices see no change in the MAC to IP address pairing, no ARP entries change or time out anywhere on the network.

If the secondary unit boots without detecting the primary unit, the secondary unit becomes the active unit and uses its own MAC addresses, because it does not know the primary unit MAC addresses. However, when the primary unit becomes available, the secondary (active) unit changes the MAC addresses to those of the primary unit, which can cause an interruption in your network traffic. Similarly, if you swap out the primary unit with new hardware, a new MAC address is used.

Virtual MAC addresses guard against this disruption because the active MAC addresses are known to the secondary unit at startup and remain the same in the case of new primary unit hardware. In multi-instance

capability the FXOS chassis autogenerates only primary MAC addresses. You can overwrite the generated MAC address with a virtual MAC address with both the primary and secondary MAC addresses, setting the secondary MAC address does ensure that to-the-box management traffic is not interrupted in the case of new secondary unit hardware.

If you do not configure virtual MAC addresses, you might need to clear the ARP tables on connected routers to restore traffic flow. The FTD does not send gratuitous ARPs for static NAT addresses when the MAC address changes, so connected routers do not learn of the MAC address change for these addresses.

The IP address and MAC address for the state link do not change at failover; the only exception is if the state link is configured on a regular data interface.

8. Select the “+” icon next to the Interface MAC Address

Interface MAC Addresses			+
Physical Interface	Active Mac Address	Standby Mac Address	
No records to display			

9. Physical Interface: GigabitEthernet0/1 Active Interface MAC Address: student choice (IP Address of interface used in example) Standby Interface Mac Address: Student Choice of input [example below]
Click Ok

NOTE*: The above step is an example of how to configure an Interface Mac Address

Add Interface Mac Address ?

Physical Interface:*
 GigabitEthernet0/1 ▼

Active Interface Mac Address:*
 0000.1981.9101

Standby Interface Mac Address:*
 0000.1981.9102

i Enter the Mac addresses in hexadecimal format such as 0123.4567.89ab

Cancel OK

10. Configure Monitored Interfaces Go to the pencil icon next to **inside** under the list of **Monitored Interfaces**

Monitored Interfaces							
Interface Name	Active IPv4	Standby IPv4	Active IPv6 - Standby IPv6	Active Link-Local IPv6	Standby Link-Local IPv6	Monitoring	
outside	198.18.133.81					●	/
diagnostic						●	/
inside	198.19.10.1					●	/

11. Enter the Standby IP Address: **198.19.10.31**. Repeat for the **outside** Interface **198.18.133.132**

Edit inside ?

☒ Monitor this interface for failures

IPv4 IPv6

Interface Name:
inside

Active IP Address:
198.19.10.1

Mask:
24

Standby IP Address:
198.19.10.31

Cancel OK

Edit outside ?

☒ Monitor this interface for failures

IPv4 IPv6

Interface Name:
outside

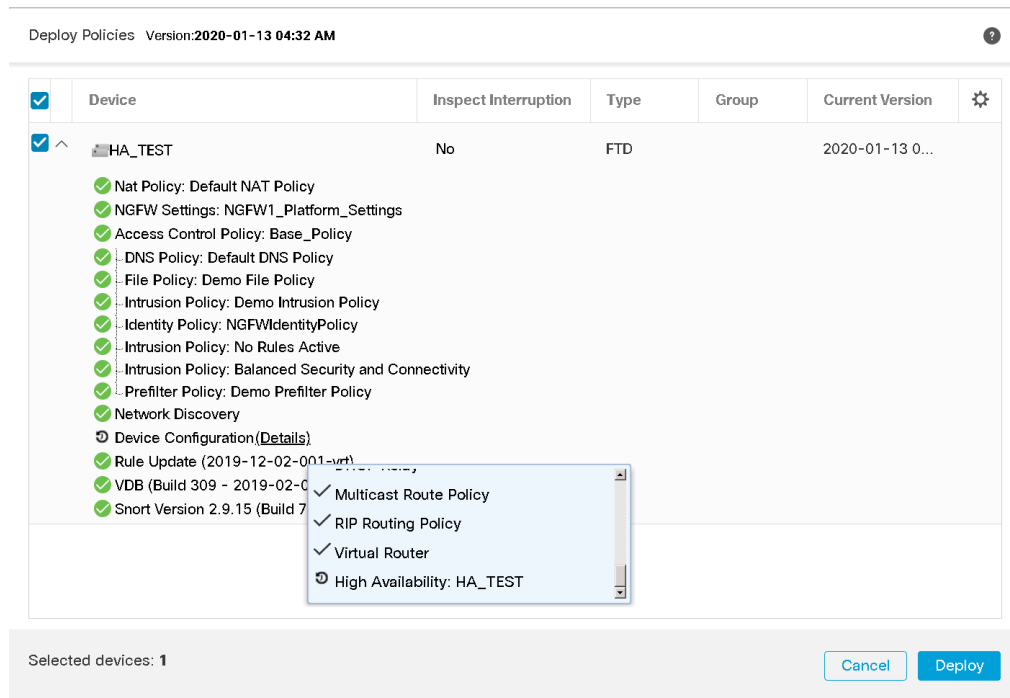
Active IP Address:
198.18.133.81

Mask:
18

Standby IP Address:
198.18.133.132

Cancel OK

12. Click **OK**, **Save** and then **Deploy** Select **HA_Test** and then Deploy

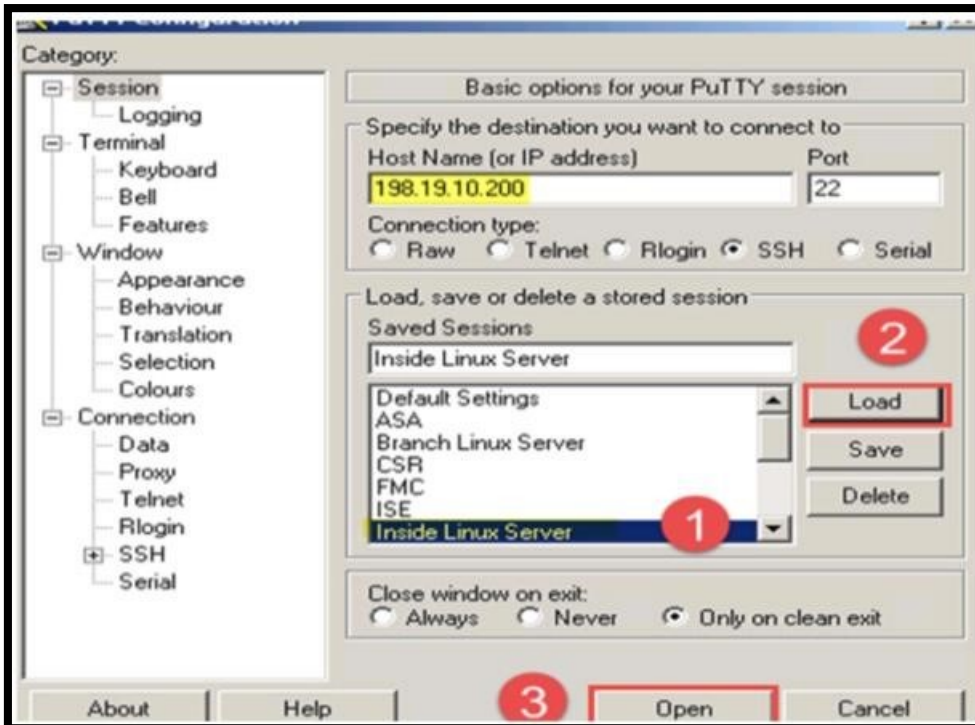


Looking at the configuration of NGFW3.

1. Let's look at some of the configuration parameters that NGFW3 received during the HA setup
2. Go to the Jump PC open PUTTY and select NGFW3
3. Login into the NGFW Username: admin Password: C1sco12345 Type:
 - a show running-config interface
 - i What is the primary IP Address of each Interface?
 - ii Is there a Standby IP Address associated with the Interface?
 - b show running-config failover
 - i What is the Failover Mac Address for Interface GigabitEthernet0/1?
 - ii What is the Interface for the Failover_Link?
 - iii What is the Interface IP Address for the Failover_Link?

Testing Failover

1. On the Jump PC go to PUTTY and open up a session to the Inside Linux Server



2. Login: **root** Password: **C1sco12345** Type: “ping outside” and let the script continue to run

```

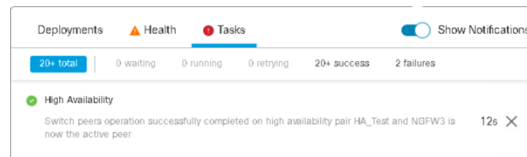
login as: root
oot@198.19.10.200's password:
ast login: Fri Dec 22 04:14:44 2017 from 198.19.10.50
root@inside ~]# ping outside
PING outside (198.18.133.200) 56(84) bytes of data.
4 bytes from outside (198.18.133.200): icmp_seq=1 ttl=63 time=33.2 ms
4 bytes from outside (198.18.133.200): icmp_seq=2 ttl=63 time=33.7 ms
4 bytes from outside (198.18.133.200): icmp_seq=3 ttl=63 time=22.5 ms
4 bytes from outside (198.18.133.200): icmp_seq=4 ttl=63 time=22.8 ms
4 bytes from outside (198.18.133.200): icmp_seq=5 ttl=63 time=23.3 ms
4 bytes from outside (198.18.133.200): icmp_seq=6 ttl=63 time=15.6 ms

```

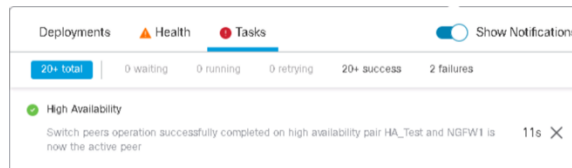
3. Go to the web interface of the FMC **Devices > Device Management** Click on the Switch Peers icon and click **Yes** when it prompts to make **NGFW3** Active.

Name	Model	Version	Chassis	Licenses	Access Control Policy	
Ungrouped (2)						
HA_TEST High Availability						
NGFW1(Primary, Active) 198.19.10.81 - Routed	FTD for VMWare	6.5.0	N/A	Base, Threat (4 more...)	Base_Policy	⚙️
NGFW3(Secondary, Active) 198.19.10.83 - Routed	FTD for VMWare	6.5.0	N/A	Base, Threat (4 more...)	Base_Policy	⚙️

4. Resize the Firefox window so you can also see the results of the pinging from the Inside Linux Server.



5. Switch back so that the **NGFW1** becomes **Active** again.



Note: The **Active** HA status of the NGFW implies that it is participating in the traffic processing and owns the Active Interface MAC addresses. Only the Primary or the Secondary NGFW can be Active at a given time and therefore handle traffic.

For the next scenarios 3, 4, 5, you will be configuring Remote Access VPN [RAVPN] with RADIUS, Multiple Factor Authentication and Enhancement using the FDM [Firepower Device Manager]. You can do all or any of the three different scenarios. Also note that the High Availability [HA] is an optional lab. The screenshots reflect a non-HA environment. If you have completed the HA scenario substitute HA_Test for NGFW1.

Scenario 3 . Monitoring and Troubleshooting

This exercise consists of the following tasks.

- Monitoring AnyConnect user activity using FMC
- Troubleshooting FDM Deployment
- Troubleshooting General Connectivity Issues across the FTD

Monitoring AnyConnect user activity

NOTE: This scenario demonstrates Troubleshooting and Monitoring Anyconnect VPN users. This section can be done only after completing the Remote VPN Scenario from the FMC VPN Lab guide.

In this section, you can monitor all active users who have logged in through AnyConnect.

1. From the Cisco dCloud Topology Map
 - a. Click on WKST 2
 - b. Login: Administrator Password: C1sco12345
 - c. Start an AnyConnect Session
 - i Either by clicking in the bottom tray on the Windows Desktop
 - ii or Start and type Anyconnect in the search bar
 - d. Click on the Connect Button [Should connect to ngfw1]
 - i Username: vpnuser
 - ii Password: C1sco12345
 - iii If you did the Multifactor Lab you will have the additional password of **push** for the second password
 - 1 confirm the Duo push
2. In the FMC, navigate to **Overview > Dashboards > Access Controlled User Statistics**
3. **Select the VPN tab.** Note that there are 7 widgets dedicated to VPN traffic.
4. Navigate to **Analysis > Users > Active Sessions.**
 - a. Notice that you see vpnuser VPN session.
 - b. **Check the box** to the left of vpnuser session and click **Logout**. When prompted, click **Continue**.

NOTE: You may also see other active sessions discovered with network discovery. For example, you may see guest discovered through an FTP session. For brevity, those sessions were left out of the figure above. If you want more details about users and how they were discovered, navigate to Analysis > Users > Users.

5. On Outside-PC, confirm that vpnuser has been logged out.
6. In the FMC, navigate to **Analysis > Users > User Activity**. In this window you will see details of current and past user sessions. Spend a couple minutes reviewing the information on this page.

Troubleshooting of AnyConnect VPN sessions

In this section, you will modify the Syslog level for VPN events on the NGFW. You will also run some basic troubleshooting commands from the NGFW1 CLI.

1. In the FMC, navigate to **Device > VPN > Troubleshooting**. Note that no records are displayed.

2. In the FMC, navigate to **Devices > Platform Settings**.
 - a. Modify Existing Platform Settings or Click New Policy **Threat Defense Settings Policy**.
 - b. Name the policy HA_Test Settings Policy.
 - c. Select the HA_Test [or NGFW1] device and **click Add to Policy**.
 - d. Click **SAVE**

Policy Assignments

Targeted Devices

Select devices to which you want to apply this policy.

Available Devices

Q Search by name or value

HA_Test
NGFWBR1

Add to Policy

Selected Devices

HA_Test

Cancel OK

3. Click **Save**. Wait for the policy to open for editing.
4. In the left navigation pane, select **Syslog**.
 - a. Under **VPN Logging Settings** change the logging level to **informational**. Note that in a production environment, it is recommended that you set this to errors or alerts.
 - b. Click **Save**.

ARP Inspection
Banner
DNS
External Authentication
Fragment Settings
HTTP
ICMP
Secure Shell
SMTP Server
SNMP
SSL
Syslog
Timeouts
Time Synchronization
UCAPL/CC Compliance

Logging Setup Logging Destinations Email Setup Event Lists Rate Limit Syslog Settings Syslog Servers

Basic Logging Settings

☐ Enable Logging

☐ Enable Logging on the failover standby unit

☐ Send syslogs in EMBLEM format

☐ Send debug messages as syslogs

Memory Size of the Internal Buffer

4096

(4096-52428800 Bytes)

VPN Logging Settings

☒ Enable Logging to FMC

Logging Level

informational

Specify FTP Server Information

☐ FTP Server Buffer Wrap

IP Address*

Username

5. Deploy the changes to the **HA_Test**
6. On the Outside-PC, generate some VPN activity. For example, connect and disconnect a VPN session.
7. In the FMC, return to **Device > VPN > Troubleshooting**. You should see records. If you do not, try adjusting the time window on this page.
8. On the **NGFW1** CLI run some of the following commands to get a rough scope of the troubleshooting capabilities. These are useful when troubleshooting RA VPN. They are primarily included for your reference.
 - a. show vpn-sessiondb ?
 - b. test aaa-server ?
 - c. debug crypto ca ? (good for trouble-shooting certificate issues)
 - d. debug crypto ipsec ?
 - e. debug ldap ?
 - f. debug aaa ?

FDM Deployment Troubleshooting

1. Open the Firefox browser click on the FDM Tab.
2. Click **Device > Routing > View Configuration**
3. Move to **Actions Column**
4. Click on the **pencil** icon on Line #1
5. Click on the **Dropdown Arrow** by Gateway

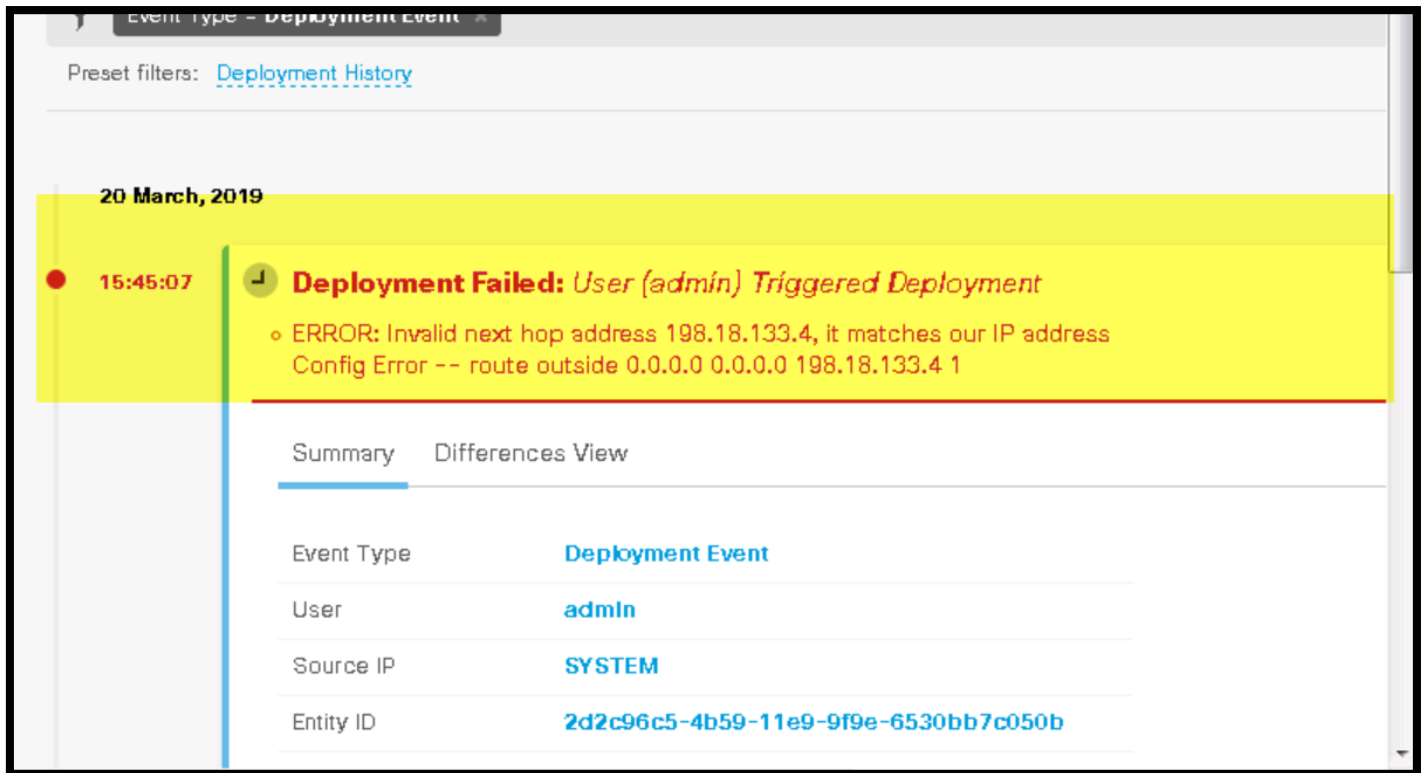
6. Select **Create New Network**
 - a. Name: **tsroute**
 - b. Host: **198.18.133.82** click **OK**
7. For Gateway select the newly created gateway: **tsroute**
8. For Interface select: **outside**
9. Click **OK**
10. Click **Deploy**
11. Wait for the Deployment to finish
12. You will see a **Status** of **Failed**
13. Click the see details

Pending Changes

✖ **Last Deployment Failed**
20 Mar, 2019 03:43:19 PM. [See Details](#)

Deployed Version	Pending Version										
⊕ Network Object Added: <i>tsroute</i> <table border="1"> <tr> <td>subType:</td> <td>Host</td> </tr> <tr> <td>value:</td> <td>198.18.133.4</td> </tr> <tr> <td>isSystemDefined:</td> <td>false</td> </tr> <tr> <td>dnsResolution:</td> <td>IPV4_AND_IPV6</td> </tr> <tr> <td>name:</td> <td>tsroute</td> </tr> </table>		subType:	Host	value:	198.18.133.4	isSystemDefined:	false	dnsResolution:	IPV4_AND_IPV6	name:	tsroute
subType:	Host										
value:	198.18.133.4										
isSystemDefined:	false										
dnsResolution:	IPV4_AND_IPV6										
name:	tsroute										
✎ Static Route Container Edited: <i>Static-Route-Entry-Container</i>											

LEGEND



14. You will see an error that references the next hop address
 - a. Go back to the FDM and fix the issue

Troubleshooting with Packet Tracer and Packet Captures

1. When to use Packet-Tracer
 - a. Verify if traffic to a specific port is allowed by the Lina Data path and Snort
 - i. Security Intelligence (IP Reputation)
 - ii. L3/L4 IPS Intrusion Rules
 - b. Packet Tracer **Does Not** currently work with: (Because it cannot emulate a L7 packet)
 - i. Identity-based rules
 - ii. L7-related (SI DNS/URL, App ID, File Policy, L7 Intrusion Rules)

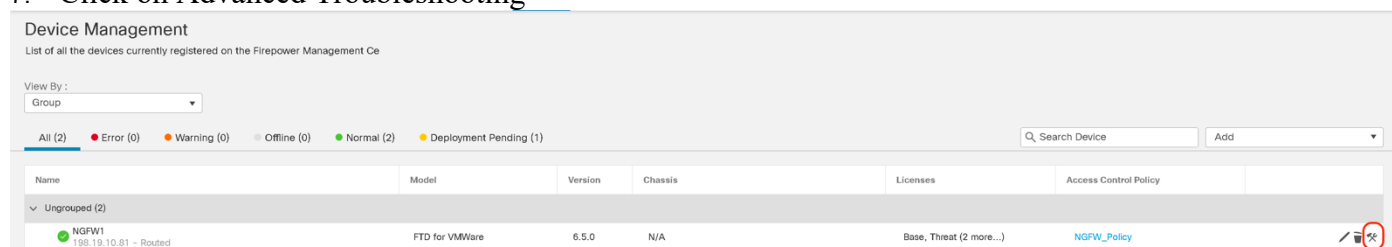
Packet-Tracer Lab

1. On the FMC go to **Policies > Access Control > Edit the Base_Policy**
2. Click Add New Rule
 - a. Name: Packet-Trace Rule
 - b. Set the Rule ABOVE rule 1
 - c. Under Action Block or Block with reset
 - d. Zones: Source Zone InZone1, Destination Zone Outzone
 - e. Networks: Source: MainOfficenetwork Destination Networks: any-ipv4

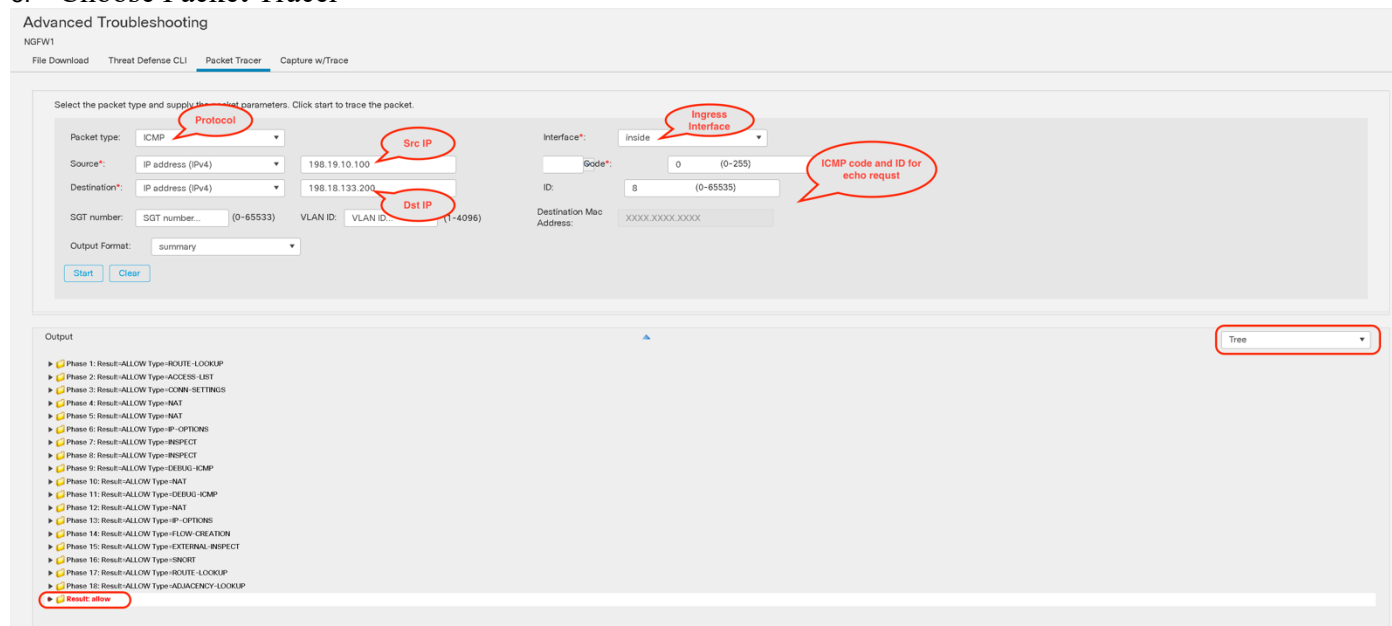
- f. Applications: Available Applications type ICMP and Select All apps matching the filter click Add to Rule
- g. Available Applications type: FTP. Select All apps matching the filter and click Add to Rule
- h. Click Logging
 - i. Click Log at Beginning of Connections
- i. Click **Add**
- j. Click **Save and Deploy** to **HA_Test** or **[NGFW1]**

NOTE: We selected all the applications related to **ICMP** and **FTP** in a production environment you would be more specific with what particular applications you are blocking.

3. Open a **PUTTY** Session to **NGFW1** Username: **admin** Password **C1sco12345**
4. Type the following **packet-tracer** input inside **icmp 198.19.10.200 8 8 198.18.133.200**
 - a. Look at Phases you will notice that the packet has been handed off to **SNORT** for further processing
 - b. You will see that **SNORT** used block w/reset a rule id to order a drop of the packet.
5. Now look at the Packet-Trace command in the FMC
6. Go to **Devices > Device Management.> NGFW1** click on the **Troubleshooting Icon**
7. Click on **Advanced Troubleshooting**



8. Choose Packet Tracer



- a. Packet Type: **ICMP**
- b. Interface: **inside**
- c. Source: **198.19.10.200**
- d. Type: **8 (Echo Request)** Code **0**
- e. Destination: **198.18.133.200**
- f. Click **Start**

NOTE: You will get the same results that you saw in the Command Line of the **NGFW1** it is just shown in the window.

9. Set up the Packet Tracer for **FTP**
 - a. Packet Type: **TCP**
 - b. Source: **198.19.10.200**
 - c. Source Port: **1111**
 - d. Destination **198.18.133.200 (Outside Linux Server)**
 - e. Destination Port: **FTP**
 - f. Click **Clear**
 - g. Click **Start**

NOTE: Phase 2 is still checking the rule you created Look at **Phase 14** you will see that SNORT looked at the rule and the verdict was to pass the packet. The first part of the packet is passed but not the next packets. To test this, go to the Jump PC and open the **inside linux server session and type ftp outside you will be prompted: login: guest** you will receive a message that states No Control connection for command Transport endpoint is not connected. You can go to **Analysis Connection Events** and see that **FTD was Blocked with reset.**

Capture w/Trace Lab

NOTE: There are two types of Traffic Captures the **Lina based** and the **Snort based**.

1. Lina Level **capture**
2. SNORT Level **capture-traffic**

Add Capture

Name*: Interface*:

Match Criteria:

Protocol*:

Source Host*: Source Network:

Destination Host*: Destination Network:

☐ SGT number: (0-65533)

Buffer:

Packet Size: 14-1522 bytes ☐ Continuous Capture ☒ Trace

Buffer Size: 1534-33554432 bytes ☒ Stop when full Trace Count:

3. Go to **Devices > Device Management** > click on the Troubleshooting Icon for NGFW1
4. Click on **Advanced Troubleshooting**
5. Click on **Capture w/Trace**
6. Click **Add Capture**
 - a. Name: Capturewtrace
 - b. Interface: inside
 - c. Protocol: ICMP
 - d. Source Host: 198.19.10.200 (Inside Linux Server)
 - e. Destination Host: any
 - f. Buffer Size: 33554432 (32 MB)
 - g. Trace Count 100
 - h. Save

NOTE: We have not removed the access policy denying **ICMP** so the pings will fail, but you will be able to see the packet shown. Also, you will export the file in PCAP format to **Wireshark** in this lab.

7. Go to the Jump PC and on the Inside Linux Server type **ping outside**.
8. If you don't see information in the Packets Shown Window in about 10 seconds hit the refresh.
9. **Once you see packets stop the ping.**
10. Click on the Save icon for the packet capture you created.
 - a. Save the file as PCAP.
11. When Prompted **Save File** and click **OK**.
12. Go to the downloads arrow of Firefox and select the file just downloaded.
13. Minimize the Browser and you will see the file opened in **Wireshark**.

14. Notice that the messages have been **administratively filtered**.

Scenario 4. Cisco Threat Intelligence Director (CTID)

This exercise consists of the following tasks.

- Upload a list of URLs to CTID that will trigger an Incident
- Subscribe CTID to a TAXII feed
- Generate CTID incidents

The CTID is a component of the FMC that can consume third party cyber threat intelligence indicators; CTID parses these indicators to produce observables that can be detected by the NGFW. The NGFW reports detection of the observables to CTID.

Then CTID determines whether the observations constitute an incident.

Steps

Two file formats are supported.

- Flat files - Lists of simple indicators such as IP addresses, URLs or SHA256 hashes.
- Threat Intelligence Director is enabled by default You can find it under **Policies > Access Control > Then the policy under Advanced.**

The screenshot shows the 'Base_Policy' configuration page. The 'Advanced' tab is selected. Under 'General Settings', the following settings are visible:

Setting	Value
Maximum URL characters to store in connection events	1024
Allow an Interactive Block to bypass blocking for (seconds)	600
Retry URL cache miss lookup	Yes
Enable Threat Intelligence Director	Yes (highlighted with a red circle)
Inspect traffic during policy apply	Yes

STIX files - XML files that can describe simple or complex indicators There are 3 ways these files can be retrieved:

- Uploaded from the computer where the FMC UI is running.
- Retrieved from a URL on a remote web server.
- Received from a TAXII feed (STIX files only).

The objective of this exercise is to configure and test CTID.

Steps

Confirm that CTID will publish observables to the NGFW

1. Navigate to **Policies > Access Control > Access Control**.
2. Edit the access control policy by clicking the pencil icon to the right of the policy.
3. Select the Advanced tab. Using this advanced setting, CTID can be enabled or disabled at the access policy level.

Base_Policy
Enter Description

Rules Security Intelligence HTTP Responses Logging **Advanced**

General Settings

Maximum URL characters to store in connection events	1024
Allow an Interactive Block to bypass blocking for (seconds)	600
Retry URL cache miss lookup	Yes
Enable Threat Intelligence Director	Yes
Inspect traffic during policy apply	Yes

4. Navigate to **Intelligence > Elements**.
5. Confirm that the NGFW1 is an element. This means that CTID can publish observables to the NGFW1 retrieved from a STIX file from a web server.

Intelligence > Elements

Name	Element Type	Registered On	Access Control Policy
NGFW1	Cisco Firepower Threat Defense for VMW...	Mar 11, 2019 1:29 AM EDT	Base_Policy
NGFWBRI	Cisco Firepower Threat Defense for VMW...	Mar 11, 2019 2:25 AM EDT	Branch Access Control Policy
NGFW3	Cisco Firepower Threat Defense for VMW...	Jan 12, 2020 10:33 PM EST	Base_Policy

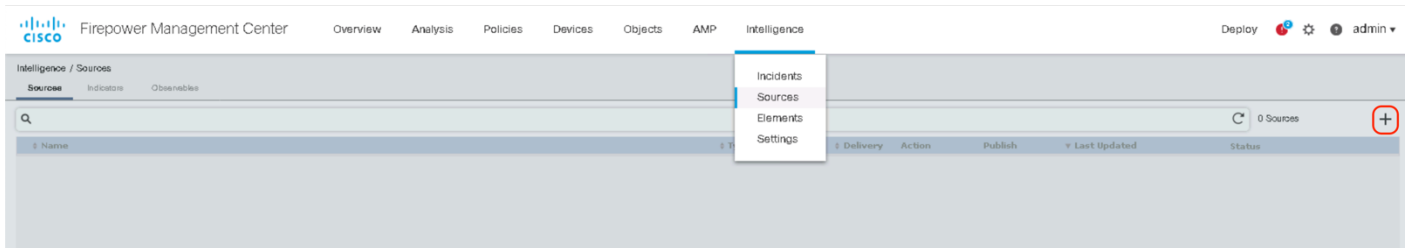
Intelligence > Settings

TID Detection
 The system is currently publishing TID observables to elements.
 Pause Resume

Incidents
Sources
Elements
Settings

NOTE: The CTID can be enabled or disabled globally. Clicking Pause will stop the CTID publishing to all elements.

6. Navigate to **Intelligence > Sources > Sources**.
7. Click the plus sign (+) on the right to add an intelligence source.



8. Upload a list of URLs to CTID that will trigger an Incident
 - a. Click the plus sign (+) on the right to add an intelligence source.
 - b. For **DELIVERY**, select **Upload**.
 - c. For **TYPE**, select **Flat File**. The **CONTENT** drop-down list will appear.
 - d. For **CONTENT**, select **URL**.
 - e. Click in the **FILE** area and select **URL_LIST.txt** from the **Files** folder on the Jump desktop.
 - f. For **NAME**, enter Local url list.
 - g. For **ACTION**, select **Block**.

Add Source

DELIVERY TAXII URL **Upload**

TYPE Flat File CONTENT URL

FILE* Drag and drop or click to attach

File attached:
URL_List.txt (90 B)

NAME* Local URL list

DESCRIPTION

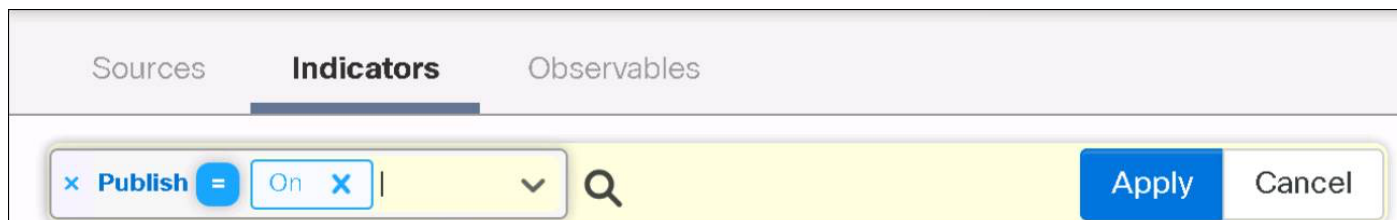
ACTION **Block**

TTL (DAYS) 90

PUBLISH ☒

Save Cancel

9. Click **Save**.
10. Wait a few seconds. Navigate to **Intelligence > Sources > Indicators**. Delete the default time-based filter. Replace it with a filter to only show published indicators, as shown in the following screenshot. Click **Apply** to apply the filter. Confirm that two URL indicators have been added.



NOTE: The incidents may not appear for a while if the original time based filter remains applied.

11. Navigate to **Intelligence > Sources > Observables**. Confirm that two type URL observables have been added.

Subscribe CTID to a TAXII feed

1. Navigate to **Intelligence > Sources > Sources**. Click the plus sign (+) on the right to add an intelligence source.
2. For **DELIVERY**, select **TAXII**.
3. For **URL**, enter <http://hailataxii.com/taxii-discovery-service>.
4. For **USERNAME**, enter **guest**.
5. For **PASSWORD**, enter **guest**.
6. For **FEEDS**, select **guest_phishtank_com**.

NOTE: It may take several seconds for the FEEDS drop-down list to populate.

7. Confirm that the screen looks like the following figure.

Add Source

DELIVERY **TAXII** URL Upload

URL* SSL Settings ▾

USERNAME

PASSWORD

⚠ Credentials will be sent using an unsecured HTTP connection

FEEDS* X ▾

Note: A separate source will be added for each feed selected. The name will default to the name of the feed and can be edited later.

ACTION

UPDATE EVERY (MINUTES) ☐ Never Update

TTL (DAYS)

PUBLISH ☒

8. Click **Save**.
9. Wait until the Status column for this source changes to **Parsing**, which may take several minutes. Do not wait for the parsing to complete - this would take too long.
10. Navigate to **Intelligence > Sources > Indicators**. Delete the default time-based filter. Replace it with a filter to only show published indicators, as you did for the previous source you configured. Click **Apply** to apply the filter. Confirm that several URL indicators have been added.
11. Navigate to **Intelligence > Sources > Observables**. Confirm that several URL observables have been added.

Generate CTID incidents

1. It can take several minutes for the observables to be published to the sensor. In this step, you will see how to confirm the publication of a particular observable. Use Putty on the Jump PC to SSH to the **NGFW1 CLI** and perform the following: (logged in as admin / C1sco12345)

1. Type **expert** to get into expert mode.
2. Type **ls -d /var/sf/*download**.

NOTE: There are several directories listed. admin@ngfw:~\$ ls -d /var/sf/*download

```
ls -d /var/sf/clamupd_download
ls -d /var/sf/iprep_download
ls -d /var/sf/sifile_download
ls -d /var/sf/cloud_download
ls -d /var/sf/sidns_download
ls -d /var/sf/siurl_download
```

Four of these (iprep_download, sidns_download, sifile_download and siurl_download) are used by security intelligence and CTID.

3. Type **grep developmentserver /var/sf/*download/*lf** (lf has a lower-case L) to confirm the observable has been published to the firewall observables list.

4. You should see a type URL CTID observable.

```
/var/sf/siurl_download/731625d4-9512-11e7-915c-
7e7252ae92ac.lf:developmentserver.com/misc/Tron.html/
```

NOTE: If you do not, wait a minute and try again. You must wait for this to be published before you go on.

On the Inside Linux server CLI:

1. Use putty on the Jump PC, to SSH to the **Inside Linux** server and perform the following : (logged in as root/C1sco12345)
 - a Run `wget -t 1 outside/files/ProjectX.pdf`. This should succeed.
 - b Run `wget -t 1 developmentserver.com/misc/Tron.html`. This should be blocked.
2. On the FMC, navigate to **Intelligence > Incidents**. Confirm that there are 2 incidents.



Last Updated	Incident ID	Indicator Name	Type	Action Taken	Status
8 minutes ago	URL-20171231-3	developmentserver.com/misc/Tron.html/	URL	Blocked	New
15 minutes ago	URL-20171231-2	developmentserver.com/misc/Tron.html/	URL	Blocked	New

3. Drill down into the incident and observe the details for this incident.
4. Confirm that there is an incident for a URL indicator. Drill down into the incident and observe the details for this incident

Appendix A. FMC Pre-configuration

After the initial installation, several configuration steps were performed on the FMC to expedite the lab exercises. These configuration steps are detailed in this appendix.

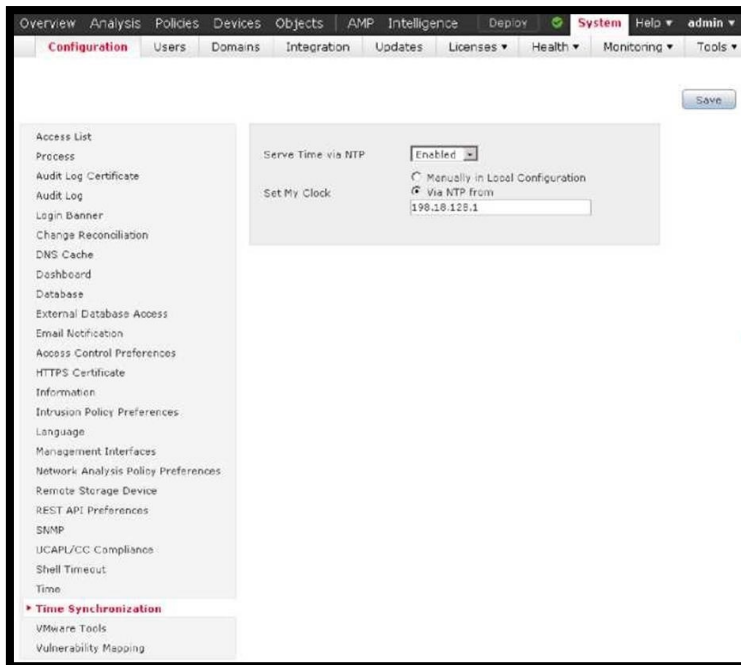
- Configuration A1,1: NTP settings
- Configuration A1,2: Demo file policy
- Configuration A1,3: Demo intrusion policy
- Configuration A1,4: Demo SSL policy
- Configuration A1,5: Custom detection list □
- Configuration A1,6: Add resetapiuser.
- Configuration A1,7: Install server certificate

- Configuration A1,7: Interface Objects

Steps

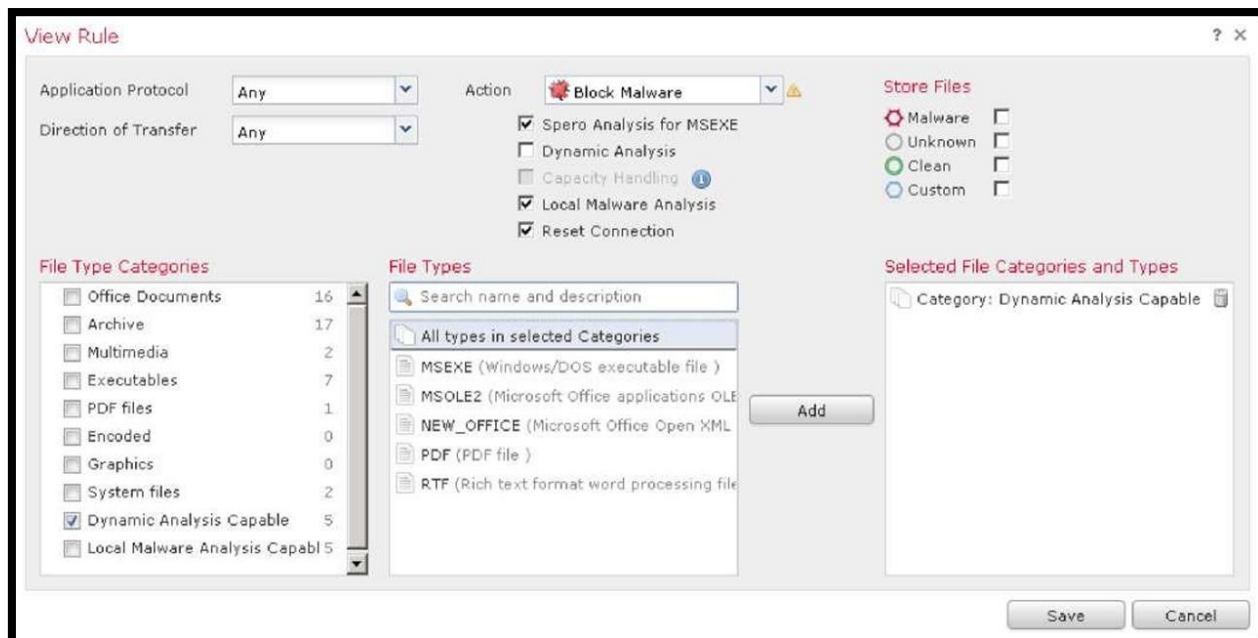
Configuration A1,1: NTP settings

1. Configure NTP settings on the FMC.
 - a. In the FMC, navigate to **System > Configuration**.
 - b. Select **Time Synchronization** from the left-side navigation pane.
 - c. Replace the default NTP server with 198.18.128.1.
 - d. Click Save.

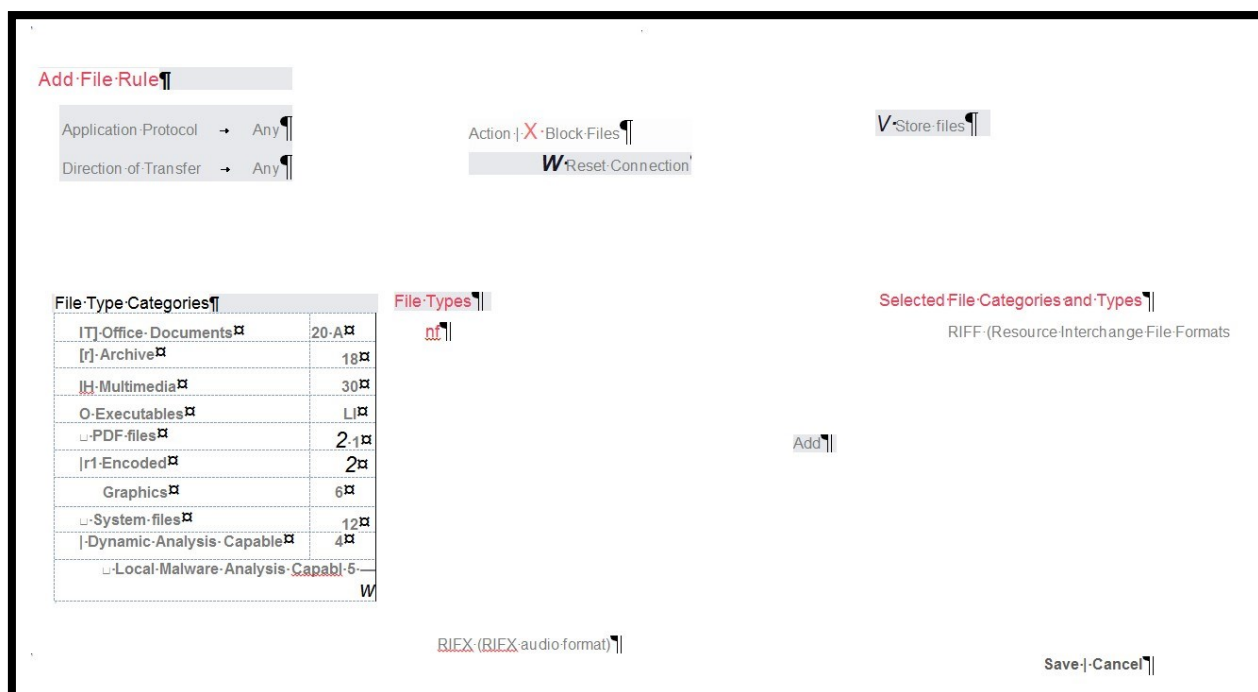


Configuration A1,2: Demo file policy

1. **Navigate to Policies > Access Control > Malware & File.**
2. **Click New File Policy. Enter a name Demo File Policy. Click Save.**
3. **Click Add File Rule.** This rule will block malware found in files MSEXEX, MSOLE2, NEW_OFFICE and PDFs.
4. **For Action select Block Malware.**
5. Check the Spero and **Local Malware Analysis** checkboxes.
6. Under **File Type Categories**, check *Dynamic Analysis Capable*. **Note** that several file types belong to this category. Click **Add**.
7. Your screen should look like the figure below.



8. Click **Save**. Ignore the warning and click **OK**, when prompted.
9. Click **Add File Rule**. This rule will block RIFF files. You will use an AVI file to test this rule, since an AVI file is a type of RIFF file. But **Note** that AVI is not listed separately as a file type.
10. **For Action select Block Files.**
11. Under **File Types**, type **riff** into the search box. Select **RIFF** from the list. Click **Add**.
12. Use default values for other settings. Your screen should look like the figure below.
13. Click **Save**.



NOTE: You cannot change the order of the rules you create. The order of the rules does not matter. The action of the rule determines its precedence. The precedence of actions is as follows.

- 1 - Block Files
- 2 - Block Malware
- 3 - Malware Cloud Lookup 4 - Detect Files
- 5 - Select the **Advanced** tab. Confirm that **Enable Custom Detection List** is selected.
- 6 - Check the **Inspect Archives** checkbox.

NOTE: Archives unable to be inspected are corrupt archive, or archives with a depth that exceeds the Max Archive Depth.

14. Click the **Save** button in the upper-right to save the file policy.

NOTE: Archives unable to be inspected are corrupt archive, or archives with a depth that exceeds the Max Archive Depth.

15. Click the **Save** button in the upper-right to save the file policy.

[Configuration A1,3: Demo intrusion policy](#)

1. **Navigate to** Objects > Intrusion Rules. **Click** Import Rules.
 - a. **Select the** Rule update or text rule file to upload and install **radio button**.
 - b. **Click** **Browse** and open the **Snort_Rules.txt** file in the **Files** folder of the Jump desktop.

NOTE: This file contains 2 simple Snort rules that are useful for testing IPS. They do not resemble published snort rules.

```
alert tcp any any -> any any (msg:"ProjectQ replaced"; content:"ProjectQ"; replace:"ProjectR"; sid:
1001001; rev:1;) alert tcp any any -> any any (msg:"ProjectZ detected"; content:"ProjectZ"; sid: 1001002;
rev:1;)
```

The first rule replaces the string ProjectQ with ProjectR. The second detects the string ProjectZ. Since the rules do not specify where the string is in the flow, they could cause issues in a production deployment.

- c. Click **Import**. The import process will take a minute or two. When it completes you will see the **Rule Update Import Log** page. Confirm that 2 rules were successfully imported.
2. **Navigate to Policies > Access Control > Intrusion.**
3. **Click Create Policy.**
 - a. **Set Name to Demo Intrusion Policy.**
 - b. Make sure that **Drop when Inline** is checked.
 - c. Select **Balanced Security and Connectivity** as **Base Policy**.

- d. **Click Create and Edit Policy.**
4. You will now modify the rules states for this new policy.
 - a. Click **Rules** under Policy Information menu on the left-hand side of the **Edit Policy** page.
 - b. Select **local** from the Category section of the rules. You should see the 2 uploaded rules. The light green arrows on the right of each rule indicate that the rules are disabled for this policy.
 - c. Check the checkbox next to the first rule. Select **Generate Events** from the Rule State drop-down menu. Click **OK**. Uncheck the checkbox next to the first rule.
 - d. Check the checkbox next to the second rule. Select **Drop and Generate Events** from the Rule State drop-down menu. Click **OK**.
 - e. Clear the filter by clicking on the X on the right side of the Filter text field.
 - f. Select **SID** from the **Rule Content** section of the rules. Enter 336 into the **Enter the SID** filter popup. Click **OK**.
 - g. Check the checkbox next to the rule. Select **Drop and Generate Events** from the **Rule State** drop-down menu. Click **OK**.

NOTE: This rule looks for a change to the root home directory in FTP traffic established on port 21. It only looks for traffic coming from the external network, but in our lab, we use the default value of \$EXTERNAL_NET, which is any, so the rule can be triggered in both directions.

An interesting exercise would be to modify this rule to search in FTP traffic in any direction, and to use the appid attribute to detect FTP traffic on any port.

Click **Policy Information** in the menu on the upper-left.

Click **Commit Changes**.

Click **OK**.

Configuration A1,4: Demo SSL policy

1. **Navigate to** Objects > Object Management > PKI > Internal CAs.
 - a. Click **Import CA**.
 - b. For **Name**, enter Verifraud.
 - c. Click the **Browse** button to the right of the text **Certificate Data or, choose a file**.
 - d. Browse to the **Certificates** folder on the Jump desktop.
 - e. **Upload** Verifraud_CA.cer.
 - f. Click the **Browse** button to the right of the text *Key or, choose a file*.
 - g. **Upload** Verifraud_CA.key.
 - h. Click **Save**.
2. You will exempt from decryption infrastructure devices, such as the FMC and AMP Private Cloud. To do this, create a network object that includes these devices.
 - a. **Navigate to** Objects > Object Management > Network.
 - b. **Click** Add Network > Add Object.
 - c. For **Name**, enter Infrastructure.
 - d. For **Network**, enter 198.19.10.80-198.19.10.130.
 - e. Click **Save** to save the network object.
3. Click the text **Add a new policy** or click the **New Policy** button.
 - a. For **Name**, enter Demo SSL Policy.
 - b. Leave the default action to **Do not decrypt**.
 - c. Click **Save**. Wait a few seconds, and the policy will open for editing.
4. Click **Add Rule**.
 - a. **For Name**, enter Exempt Infrastructure.
 - b. **Leave Action set to** Do Not decrypt.
 - c. **In the Networks tab, under Networks, select** Infrastructure, **and click** Add to Source.
 - d. Click **Add** to add this rule to the SSL policy.
5. Click **Add Rule**.
 - a. **For Name**, enter Decrypt Search Engines.

- b. **Set Action to Decrypt - Resign.**
- c. Select **Verifraud** from the drop-down list to the right of the word **with**.
- d. In the **Applications** tab, under **Application Filters**, search for **Sear**. You will see **Search Engine** under **Categories**. Check this checkbox and click **Add to Rule**.
- e. Select the **Logging** tab and check the **Log at End of Connection** checkbox.
- f. Click **Add** to add this rule to the SSL policy.
- 6. Click **Add Rule**.
 - a. For Name, enter **Decrypt Other**.
 - b. **Set Action to Decrypt - Resign.**
 - c. Select **Verifraud** from the drop-down list to the right of the word **with**.
 - d. Select the **Logging** tab and check the **Log at End of Connection** checkbox.
 - e. Click **Add** to add this rule to the SSL policy.
- 7. Click **Save** to save the SSL policy.

NOTE: The Replace Key checkbox deserves explanation. Whenever the action is set to Decrypt - Resign, Firepower will replace the public key. The Replace Key checkbox determines how the decrypt action is applied to self-signed server certificates.

If Replace Key is deselected, self-signed certificates are treated like any other server certificates. Firepower replaces the key and resigns the certificate. Generally, the endpoint is configured to trust Firepower, and therefore will trust this resigned certificate.

If Replace Key is selected, self-signed certificates are treated differently. Firepower replaces the key, and generates a new self- signed cert. The browser on the endpoint will generate a certificate warning.

In other words, checking the Replace Key checkbox makes the resign action preserve lack-of-trust for self-signed certificates.

Configuration A1,5: Custom detection list

There is a harmless file called **Zombies.pdf** that will trigger a malware event, assuming the cloud lookup succeeds. Sometimes labs have issues with cloud connectivity. Therefore, this is added to the custom detection list to ensure it will trigger a malware event.

- 1. **Navigate to Objects > Object Management > File List.**
- 2. Click the pencil icon to edit the **Custom-Detection-List**.
 - a. Select **Calculate SHA** from the **Add by** drop-down list.
 - b. Click **Browse**.
 - c. Browse to the Files folder on the Jump desktop.
 - d. Select **Zombies.pdf** and click **OK**.

- e. **Click** Calculate and Add SHAs.
- f. **Click Save.**

Configuration A1,6: Add restapiuser

It is convenient to have a separate user to use the API Explorer. This allows use of both the FMC and API Explorer at the same time.

1. **Navigate to** System > Users. **Click** Create User.
 - a. For **Username**, enter restapiuser.
 - b. For **Password**, enter **C1sco12345** Confirm the password.
 - c. Set Maximum Number of Failed Logins to 0.
 - d. Check the **Administrator** checkbox.

Configuration A1,7: Install server certificate

By default, the FMC UI uses a self-signed certificate. This is replaced by a certificate signed by the pod AD server, which the Jump browsers trust.

1. **Navigate to** Objects > Object Management > PKI > Trusted CAs.
 - a. **Click** Add Trusted CA.
 - b. For **Name**, enter **dCloud**.
 - c. **Click** the **Browse** button to the right of the text **Certificate Data or, choose a file**.
 - d. Browse to the **Certificates\Lab Certificates\Other Certificates** folder on the Jump desktop.
 - e. **Upload** AD-ROOT-CA-CERT.cer.
 - f. **Click Save.**

2. Connect to the FMC CLI via SSH. Become root by typing **sudo -i**. The Sudo password is **C1sco12345**
 - a. **Type** `cd /etc/ssl` **and then type** `cp server* /root`.
 - b. **Type** `cat > /etc/ssl/server.crt`
 - c. From the **Certificates** folder on the Jump desktop edit the file **fmc.cer** with Notepad++.
 - d. Select all, and then copy and paste into the FMC CLI
 - e. **Type** **Ctrl+D**.
 - f. **Type** `cat > /etc/ssl/server.key`
 - g. From the **Certificates** folder on the Jump desktop edit the file **fmc.key** with Notepad++.
 - h. Select all, and then copy and paste into the FMC CLI
 - i. **Type** **Ctrl+D**.
 - j. **Type** `pmttool restartbyid httpsd`.

Appendix B. REST API Scripts

Here are the two Python scripts that were used in the first lab exercise. You only run the first script **register_config.py**. It will call the second script **connect.py**, which will create the compiled file **connect.pyc**.

Python script register_config.py

```
#!/usr/bin/python import json import connect import sys host = "fmc.example.com"
username = "restapiuser" password = "C1sco12345" name="NGFW"
#connect to the FMC API headers,uuid,server = connect.connect (host, username, password) user_input
= str(raw_input("Would you like to register the managed device? [y/n]")) if user_input == "y":
policy_name = str(raw_input("Enter name of new Access Control Policy to be create:")) access_policy = {
"type": "AccessPolicy",
"name": policy_name,
"defaultAction": { "action": "BLOCK" }
} post_response = connect.accesspolicyPOST(headers,uuid,server,access_policy)
policy_id = post_response["id"] print "\n\nAccess Control Policy\n" + policy_name +
"\ncreated\n\n" device_post = { "name": name,
"hostName": "ngfw.example.com",
"regKey": "C1sco12345",
"type": "Device",
"license_caps": [
"BASE",
"MALWARE",
"URLFilter",
"THREAT"
],
"accessPolicy": {
"id": policy_id,
"type": "AccessPolicy"
} } post_data = json.dumps(device_post) output = connect.devicePOST (headers, uuid, server,
post_data) print "\n\nPost request is: \n" + json.dumps(output,indent=4) + "\n\n" GET ALL THE
DEVICES AND THEIR corresponding interfaces user_input = str(raw_input("In the FMC UI, confirm that
the device discovery has completed and then press 'y' to continue or 'n' to exit. [y/n]"))
headers,uuid,server = connect.connect (host, username, password) if
user_input == "n": quit()
devices = connect.deviceGET(headers,uuid,server) for device in devices["items"]: if device["name"]
== name: print "DEVICE FOUND, setting ID" device_id = device["id"] NOW THAT WE HAVE THE DEVICE ID WE
NEED TO GET ALL THE INTERFACES interfaces = connect.interfaceGET(headers,uuid,server,device_id)
Interfaces i want to change interface_1 = "GigabitEthernet0/0" interface_2 =
"GigabitEthernet0/1" for interface in interfaces["items"]: if interface["name"] == interface_1:
interface_1_id = interface["id"] print "interface 1 found" if interface["name"] == interface_2:
interface_2_id = interface["id"] print "interface 2 found" user_input = str(raw_input("Would you
like to configure device interfaces? [y/n]")) if user_input == "y": interface_put = {
"type": "PhysicalInterface",
"hardware": {
```

```

"duplex": "AUTO",
"speed": "AUTO"
},
"enabled": True,
"MTU": 1500,
"managementOnly": False,
"ifname": "outside",
"enableAntiSpoofing": False,
"name": "GigabitEthernet0/0",
"id": interface_1_id,
"ipv4" : {
"static": {
"address": "198.18.133.2",
"netmask": "18"
}
} } put_data = json.dumps(interface_put) connect.interfacePUT (headers, uuid, server,
put_data, device_id, interface_1_id) interface_put = {
"type": "PhysicalInterface",
"hardware": {
"duplex": "AUTO",
"speed": "AUTO"
},
"enabled": True,
"MTU": 1500,
"managementOnly": False,
"ifname": "inside", "enableAntiSpoofing": False,
"name": "GigabitEthernet0/1",
"id": interface_2_id,
"ipv4" : {
"static": {
"address": "198.19.10.1",
"netmask": "24"
}
} } put_data = json.dumps(interface_put) connect.interfacePUT (headers, uuid,
server, put_data, device_id, interface_2_id)

```

Python script connect.py

```

#!/usr/bin/python import json import sys import requests #Suppress
HTTPS insecure errors for cleaner output from
requests.packages.urllib3.exceptions import InsecureRequestWarning
requests.packages.urllib3.disable_warnings(InsecureRequestWarning)
#define function to connect to the FMC API and generate authentication token def connect (host,
username, password): headers = {'Content-Type': 'application/json'} path =
"/api/fmc_platform/v1/auth/generatetoken" server = "https://" + host url = server + path try:
r = requests.post(url, headers=headers, auth=requests.auth.HTTPBasicAuth(username,password),
verify=False) auth_headers = r.headers token = auth_headers.get('X-auth-access-token',
default=None) uuid = auth_headers.get('DOMAIN UUID', default=None) if token == None:
print("No Token found, I'll be back terminating....") sys.exit()
except Exception as err:
print ("Error in generating token --> " + str(err)) sys.exit() headers['X-auth-access-token']
= token return headers,uuid,server

```

```

def devicePOST (headers, uuid, server, post_data): api_path= "/api/fmc_config/v1/domain/" + uuid +
"/devices/devicerecords url = server+api_path try:
r = requests.post(url, data=post_data, headers=headers, verify=False) status_code = r.status_code resp
= r.text json_response = json.loads(resp) print("status code is: " + str(status_code)) if status_code ==
201 or status_code == 202: print("Post was sucessfull...") else:
r.raise_for_status() print("error ocured
in POST -->" + resp) except
requests.exceptions.HTTPError as err: print
("Error in connection --> " + str(err))
finally:
if r: r.close() return json_response def deviceGET (headers, uuid, server): api_path=
"/api/fmc_config/v1/domain/" + uuid + "/devices/devicerecords" url = server+api_path try: r =
requests.get(url, headers=headers, verify=False) status_code = r.status_code resp = r.text
json_response = json.loads(resp) print("status code is: " + str(status_code)) if status_code ==
200: print("GET was sucessfull...") else:
r.raise_for_status() print("error ocured
in POST -->" + resp) except
requests.exceptions.HTTPError as err: print
("Error in connection --> " + str(err))
finally:
if r: r.close() return json_response def
interfaceGET (headers, uuid, server, device_id):
api_path= "/api/fmc_config/v1/domain/" + uuid + "/devices/devicerecords/" + device
id + "/physicalinterfaces" url = server+api_path try:
r = requests.get(url, headers=headers, verify=False) status_code = r.status_code resp = r.text
json_response = json.loads(resp) print("status code is: " + str(status_code)) if status_code == 200:
print("GET was sucessfull...") else:
r.raise_for_status() print("error ocured
in POST -->" + resp) except
requests.exceptions.HTTPError as err: print
("Error in connection --> " + str(err))
finally:
if r: r.close() return json_response def interfacePUT (headers, uuid,
server, put_data, device_id, interface_id):
api_path= "/api/fmc_config/v1/domain/" + uuid +
"/devices/devicerecords/" + device_id + "/physicalinterfaces/" + interface_id url
= server+api_path try:
r = requests.put(url, data=put_data, headers=headers, verify=False) status_code = r.status_code resp
= r.text json_response = json.loads(resp) print("status code is: " + str(status_code)) if status_code
== 200 : print("Put was sucessfull...") else:
r.raise_for_status()
print("error ocured in POST -->" + resp) except
requests.exceptions.HTTPError as err: print
("Error in connection --> " + str(err)) finally:
if r: r.close() return json_response def
accesspolicyPOST (headers, uuid, server, post_data):
api_path= "/api/fmc_config/v1/domain/" + uuid +
"/policy/accesspolicies" url = server+api_path try:

```

```
r = requests.post(url, data=json.dumps(post_data), headers=headers, verify=False) status_code =
r.status_code resp = r.text json_response = json.loads(resp) print("status code is: "+
str(status_code)) if status_code == 201 or status_code == 202: print("Post was sucessfull...") else:
r.raise_for_status() print("error occured in POST -->" +resp) except
requests.exceptions.HTTPError as err: print ("Error in connection --> "+str(err))
finally:
if r: r.close() return json_response
```



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